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Beyond GDP: The Quest for a Measure of Social Welfare

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This paper critically examines the various approaches to the measurement of individual well-being and social welfare that have been considered for the construction of alternatives to GDP. Special attention is devoted to recent developments in the analysis of sustainability, in the study of happiness, in the theory of social choice and fair allocation, and in the capability approach. It is suggested in the conclusion that, although convergence toward a consensual approach is not impossible, for the moment not one but three alternatives to GDP are worth developing. (JEL I31, E23, E01)

1. Introduction

GDP is recurrently criticized for being a poor indicator of social welfare and, therefore, leading governments astray in their assessment of economic policies. As is well known, GDP statistics measure current economic activity but ignore wealth variation, international income flows, household production of services, destruction of the natural environment, and many determinants of well-being such as the quality of social relations, economic security and personal safety, health, and longevity. Even worse, GDP increases when convivial reciprocity is replaced by anonymous market relations and when rising crime, pollution, catastrophes, or health hazards trigger

defensive or repair expenditures. Not surprisingly, the construction of better indicators of social welfare is also, recurrently, a hot issue in public debate and a concern for politicians and governments. The last two decades have witnessed an explosion in the number of alternative indicators and a surge of initiatives from important institutions such as the OECD, the UNDP, the European Union—more recently the French government has appointed a committee, chaired by Joseph E. Stiglitz and including four other Nobel Prize winners, to propose new indicators of “economic performance and social progress.”

In the meantime, welfare economics¹ has burgeoned in various directions, involving the theory of social choice, the theory of

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¹ The expression “welfare economics” is used here in a very broad sense, including all branches of economics that bear on the definition of criteria for the evaluation of social states and public policies. It is not restricted to the narrow confines of Old and New (or New New) Welfare Economics.

fair allocation, the capability approach, the study of happiness and its determinants, in conjunction with new developments in the philosophy of social justice and the psychology of well-being. These conceptual developments provide new analytical tools that may be directly useful for concrete measurements. About a decade ago, Daniel T. Slesnick (1998) made the following observation: "While centrally important to many problems of economic analysis, confusion persists concerning the relationship between commonly used welfare indicators and well-established theoretical formulations" (p. 2108). It is probably safe to say that much the same now holds about the relationship between concrete measures of welfare—old, new, and potential—and up-to-date theories. It appears timely to ask what the existing academic literature has to say about alternatives to GDP.

The practical importance of a measure of social welfare can hardly be overstated. Policy decisions, cost–benefit analyses, international comparisons, measures of growth, and inequality studies constantly refer to evaluations of individual and collective well-being. The fact that monetary measures still predominate in all such contexts is usually interpreted as imposed by the lack of a better index rather than reflecting a positive consensus.

The purpose of this paper is, in the light of state-of-the-art welfare economics, to examine the pros and cons of the main alternative approaches to the measurement of social welfare from the perspective of policy evaluation as well as international and intertemporal comparisons. Four approaches are discussed here. First, there is the idea of a "*corrected GDP*" that would take account, in particular, of nonmarket aspects of well-being and of sustainability concerns. As will be explained here, a basic problem for this approach is that its starting point, national income, as a candidate for a measure of social welfare,

is much less supported by economic theory than is commonly assumed. The extension of this approach to intertemporal welfare as attempted in "green" accounting adds even more complications. In view of recent developments in the theory of social choice and fairness, it will be argued that the idea of a "corrected GDP" is still defensible but implies different accounting methods than usually thought. Second, there is the idea of "*Gross National Happiness*," which has been revived by the burgeoning happiness studies. It will be argued here that the happiness revolution might, instead of bringing about the return of "utility," ultimately condemn this concept for being simplistic, and reveal that subjective well-being cannot serve as a metric for social evaluation without serious precautions. Third, there is the "*capability approach*" proposed by Amartya Sen, primarily as a framework for thinking rather than a precise method of measurement. This approach has now inspired a variety of applications, but most of its promoters are reluctant to seek a synthetic index, a famous exception being the Human Development Index (HDI). It will be argued here that a key aspect of this problem is whether individual valuations of the relevant dimensions of capability can and should be taken into account—an issue over which a dialogue with the two previous approaches might prove very useful. Fourth, there is the growing number of "*synthetic indicators*" that, following the lead of the HDI, are constructed as weighted averages of summary measures of social performance in various domains. It will be argued here that, if the three other approaches were fully exploited, there would be little reason to keep this fourth approach alive because it is ill-equipped to take account of the distribution of well-being and advantage among the members of society.

The paper is structured as follows. Sections 2–4 deal with monetary measures that are linked to the project of a corrected

GDP. Section 2 revisits the classical results involving the value of total consumption and usually invoked in justification of GDP-like measures. This appears important because some of these results are often exaggerated, while others are little known or even susceptible of developments in future research. Section 3 is devoted to the intertemporal extension of this approach, as featured in the Net National Product (NNP) and “green” accounting. Section 4 turns to measures based on willingness-to-pay and money-metric utilities, highlighting the connection with recent developments in the theory of social choice and fairness. This section also briefly discusses cost–benefit analysis, which is an important tool for policy evaluation. Sections 5–7 are devoted to the nonmonetary approaches, namely, synthetic indicators such as the HDI (section 5), happiness studies and the various possible indexes of subjective well-being (section 6), and the capability approach (section 7). Section 8 makes concluding remarks about the relative strengths and weaknesses of the various approaches analyzed in the paper and the prospects for future developments and applications.

2. Monetary Aggregates Revisited

The project of correcting GDP has been often understood, after William D. Nordhaus and James Tobin’s (1973) seminal work, as adding or subtracting terms that have the same structure as GDP, i.e., monetary aggregates computed as quantities valued at market prices or at imputed prices in case market prices are not available.² As we will see in this section, economic theory is much less supportive of this approach than usually

thought by most users of national accounts. Many official reports swiftly gloss over the fact that economic theory has established total income as a good index of social welfare under some assumptions (which are usually left unspecified). To be sure, there is a venerable tradition of economic theory that seeks to relate social welfare to the value of total income or total consumption.³ Most of that theory, however, deals with the limited issue of determining the sign of the welfare change rather than its magnitude, not to mention the level of welfare itself. In this perspective, the widespread use of GDP per capita, corrected or uncorrected, as a cardinal measure allowing percentage scaling of differences and variations appears problematic.⁴ In this section, I review the old and recent arguments for and against monetary aggregates as social welfare indicators.

2.1 A Revealed Preference Argument

Start from the revealed preference argument that, assuming local nonsatiation, if a consumer chooses a commodity bundle x (with ℓ different commodities) in a budget set defined by the price vector p , then x is revealed preferred to all bundles y such that $py < px$. If x is interior and assuming differentiability, for an infinitesimal change dx , $x + dx$ is strictly preferred to x by the consumer if and only if $pdx > 0$. Note the importance of the interiority assumption here. If the consumer does not consume some of the goods, $pdx > 0$ is compatible with a decrease in satisfaction. To simplify matters, we suppose

³ Earlier surveys can be found in Sen (1979) and Slesnick (1998).

⁴ Although welfare economists have often defined their task as finding an ordinal ranking of social welfare rather than a quantitative index, most people seem to think otherwise, as illustrated by the way GDP is used. Obviously, it would be valuable to have a meaningful scale of social welfare, and several indicators discussed in this paper (equivalent income, U -index, HDI) have this property.

² Nordhaus and Tobin (1973) set out to compute “a comprehensive measure of the annual real consumption of households. Consumption is intended to include all goods and services, marketed or not, valued at market prices or at their equivalent in opportunity costs to consumers” (p. 24).

in this section that commodity bundles are elements of $\mathbb{R}_+^\ell$.

The revealed preference argument can be extended to social welfare $W(u_1(x_1), \dots, u_n(x_n))$ and total consumption $X = x_1 + \dots + x_n$ under various sets of assumptions. One approach consists in considering a representative agent who consumes X and whose utility $U(X)$ is identified with social welfare. This approach is quite discredited. Even when the aggregate demand is consistent with the behavior of a representative agent, there is little reason to identify the representative agent's revealed preferences with social welfare. As stressed in Kirman (1992), the representative agent can appear to be better off even when all individual members of the population are actually made worse off.⁵

Another approach, due to Paul A. Samuelson (1956), consists in assuming that X is optimally distributed, in the sense that $(x_1, \dots, x_n)$ maximizes $W(u_1(x_1), \dots, u_n(x_n))$ subject to the constraint $x_1 + \dots + x_n = X$. Under this assumption there is a welfare function that depends on total consumption:

$$(1) \quad W^*(X) = \max W(u_1(x_1), \dots, u_n(x_n))$$

$$\text{s.t. } x_1 + \dots + x_n = X.$$

It is much like a utility function and can serve to apply the revealed preference argument to total consumption for the evaluation of market allocations. The limitation of this approach is that it is not realistic to assume that consumption or wealth can be redistributed at will by lump-sum transfers (as is implicit in positing that the only constraint is $x_1 + \dots + x_n = X$), and that the status quo is socially optimal. Dropping either of these two assumptions invalidates the approach. When

the status quo is not optimal, an improvement in social welfare is even compatible, obviously, with a decrease in total consumption, not just a decrease in the value of total consumption. When redistribution is made by distortionary taxes, at a second-best optimum the social marginal value of income for different individuals typically remains unequal, so that a change of allocation that favors individuals with greater social marginal value may improve social welfare even if the market value of total consumption is reduced.

2.2 Decomposing Welfare

A third approach consists in decomposing social welfare into a sum and a distributive component. Focusing on the sum can then be justified as legitimate under the explicit proviso that one ignores the distributional aspect of social welfare. Consider a simple social welfare function defined directly on expenditures or incomes, $W(m_1, \dots, m_n)$, and assumed to be increasing in each argument. Following Serge-Christophe Kolm (1969) and Anthony B. Atkinson (1970), one may compute the equally distributed equivalent $e(m_1, \dots, m_n)$, defined as the amount of income which, equally given to all individuals, yields the same social welfare as the distribution $(m_1, \dots, m_n)$. Formally, $e(m_1, \dots, m_n)$ is equal to the solution x to the equation:

$$W(x, \dots, x) = W(m_1, \dots, m_n).$$

As the solution x increases with the value of $W(m_1, \dots, m_n)$, the function $e(m_1, \dots, m_n)$ is ordinally equivalent to $W(m_1, \dots, m_n)$. For inequality averse social preferences, $e(m_1, \dots, m_n)$ is less than the average income. This makes it possible to write the decomposition

$$(2) \quad e(m_1, \dots, m_n) = \frac{1}{n} \sum_i m_i \times \frac{e(m_1, \dots, m_n)}{\frac{1}{n} \sum_i m_i},$$

⁵ Intuitively, because the representative agent compares much larger bundles than the agents, it may, with quite different preferences, have a demand that mimics the sum of individual demands over a large range of prices.

in which the second term captures distributional preferences and the first term is, as desired, a multiple of total income. This kind of decomposition can be generalized but if one wants to have a term representing total or average income or expenditure, the argument of the function W must be the vector of incomes or expenditures, which is rather restrictive.⁶

There are other ways to make total expenditures appear in a decomposition. Slesnick (1998) proposes an additive decomposition of social welfare based on Robert A. Pollak's (1981) social expenditure function. This function is defined as follows, with a reference price vector p^* :

$$(3) \quad V(x_1, \dots, x_n) = \min p^*(y_1 + \dots + y_n) \\ \text{s.t. } W(u_1(y_1), \dots, u_n(y_n)) \geq W(u_1(x_1), \dots, u_n(x_n)).$$

Under mild regularity conditions, this function is well defined and is ordinally equivalent to $W(u_1(x_1), \dots, u_n(x_n))$. The first term in Slesnick's decomposition is the value of Pollak's social expenditure function at p^* for the maximum level of social welfare that could be obtained by redistributing the total income at $(x_1, \dots, x_n)$ at current prices. If $(x_1, \dots, x_n)$ is a market allocation with current prices p^* , this equals p^*X . The second term is the difference between $V(x_1, \dots, x_n)$ and the first term. Dale W. Jorgenson (1990) makes a similar decomposition in a multiplicative fashion—the first term is total expenditure and the second term is the ratio of Pollak's social expenditure function to total expenditure:

$$(4) \quad V(x_1, \dots, x_n) = p^*X \times \frac{V(x_1, \dots, x_n)}{p^*X}.$$

⁶ This respects individual preferences only if prices are fixed (or at least relative prices, if income is deflated), and aversion to income inequality is defensible only if individuals do not have substantially unequal needs.

A related kind of decomposition has been proposed earlier by J. de V. Graaff (1977). He computes an efficiency index as the smallest fraction of any producible bundle that maintains everyone at current satisfaction,⁷ normalizes social welfare as the smallest fraction of any producible bundle that maintains social welfare at its current level, and measures an equity index as the ratio of normalized social welfare over efficiency. Then normalized social welfare is the product of the efficiency and equity indexes. Graaff's efficiency term is based on the Scitovsky frontier and one may regret that a similar measure of the efficiency of the current allocation is absent from the decompositions proposed by Slesnick and Jorgenson—in effect, they restrict attention to efficient allocations. On the other hand, Graaff's decomposition does not feature total expenditure and his measure of normalized social welfare is not invariant to changes in production possibilities, unlike Pollak's social expenditure function. This is an area where research could still make progress.

For small variations, assuming differentiability an altogether different (and simpler) decomposition of social welfare is obtained by manipulating the formula

$$dW(u_1(x_1), \dots, u_n(x_n)) = \sum_{i=1}^n \frac{\partial W}{\partial u_i} du_i \\ = \sum_{i=1}^n \beta_i p dx_i,$$

⁷ This bears some similarity with Gerard Debreu's (1951) coefficient of resource utilization (which is the smallest fraction of the resources available before production that would have made it possible, given the technology and the preferences, to maintain everyone at current satisfaction—unlike Graaf's measure, this coefficient depends on the role of commodities as net inputs or net outputs in production).

where $\beta_i = (\partial W / \partial u_i) du_i / p dx_i$, so as to get (letting $\bar{\beta}$ denote the mean $(1/n) \sum_{i=1}^n \beta_i$):⁸

$$(5) \quad dW(u_1(x_1), \dots, u_n(x_n)) \\ = \bar{\beta} p dX + \sum_{i=1}^n (\beta_i - \bar{\beta})(p dx_i - p dX/n).$$

Although this formula is of a purely accounting sort—it involves no behavioral assumption—and holds for any arbitrary price vector, it is most intuitive to read it while thinking of an initial market allocation with price vector p . For an interior market allocation, β_i equals the social marginal value of money for i , i.e., the maximum increase in W that can be obtained by spending an additional dollar on i .⁹ In general, β_i is simply the actual marginal social return on the incremental money spent on i (it can even be negative if $du_i < 0$ while $p dx_i > 0$, as can happen in a noninterior allocation). The second term in the decomposition depends on the correlation between the social priority of individuals and their consumption variation and is quite intuitive for a distributional term.¹⁰

Do formulae like (4) and (5) make it legitimate to focus on the evolution of total expenditure provided it is explicitly stated that distributive issues are ignored? The fact that both formulae can be computed equally well for arbitrary price vectors suggests that one should not make too much of such decompositions. Bad news can often

be described as the combination of an even greater catastrophe with some good event—such manipulation seldom gives a serious reason to rejoice. On the other hand, an analysis of the evolution of social welfare based on such decompositions should obviously appear as useful information *provided that* all the terms of the decomposition are computed and displayed. In this light, standard aggregate income or expenditure figures are (no more, but no less than) a relevant part of the full picture.

2.3 A Variant of the Revealed Preference Argument

Sen (1976) considers a variant of the revealed preference argument. If consumer preferences are convex, at any bundle x there is a price vector p such that for all bundles y , $py < px$ implies that x is preferred to y . This argument does not assume that x maximizes utility in the budget set, which means that the vector p need not coincide with market prices. In fact some components of x may not even be marketed.

As explained by Sen, this argument can be immediately extended to social welfare if social welfare is measured by a Bergson-Samuelson function $\bar{W}(x_1, \dots, x_n)$ —defined directly on quantities—that is quasi-concave. But the corresponding price vector $\bar{p}$ must then have $n\ell$ components. If $(x_1, \dots, x_n)$ is a market allocation in which all individuals face the same ℓ -price vector p , then the $n\ell$ -vector $\bar{p}$ can be chosen such that the ratios $\bar{p}_{ik}/\bar{p}_{il}$ (individual i , commodities k and l) are the same for all i . The computation of the inner product $\bar{p}(x_1, \dots, x_n)$ then simply boils down to $\alpha_1 p x_1 + \dots + \alpha_n p x_n$, for a vector of weights $(\alpha_1, \dots, \alpha_n)$ that embodies distributional preferences. If these weights are scaled so that $\alpha_1 + \dots + \alpha_n = n$, the sum can be decomposed into two terms:

$$(6) \quad pX \times \frac{\alpha_1 p x_1 + \dots + \alpha_n p x_n}{pX},$$

⁸ The formula uses the fact that the mean of a product equals the product of the means plus the covariance.

⁹ This is because, in this case, the ratio $(\partial u_i / \partial x_{ik}) / p_k$ does not depend on k , so that for all k ,

$$\beta_i = \frac{\partial W}{\partial u_i} \frac{\sum_{k=1}^{\ell} \frac{\partial u_i}{\partial x_{ik}} dx_{ik}}{p dx_i} = \frac{\partial W}{\partial u_i} \frac{\partial u_i}{\partial x_{ik}} \frac{1}{p_k}.$$

¹⁰ It should not be confused with the “distributional characteristic” (Martin S. Feldstein 1972a, 1972b; Atkinson and Stiglitz 1980) that plays an important role in the analysis of optimal indirect tax or public pricing. The distributional characteristic depends on the correlation between β_i and the level rather than the variation of consumption of a commodity.

the second of which is less than unity if $(\alpha_1, \dots, \alpha_n)$ and $(px_1, \dots, px_n)$ are inversely ranked, i.e., in case of inequality aversion over expenditures. For instance, if the weights are simply proportional to the ranks (giving rank 1 to the richest, and so on) of the individuals in $(px_1, \dots, px_n)$, Sen shows that, for a large n ,

$$(7) \quad \frac{\alpha_1 px_1 + \dots + \alpha_n px_n}{pX} \simeq 1 - G(px_1, \dots, px_n),$$

where $G(px_1, \dots, px_n)$ is the Gini coefficient.¹¹

2.4 Limitations of the Approach

A noteworthy feature of the approaches described in this section is their silence about how to make interpersonal comparisons—with two exceptions: the income approach in (2) and the rank-order weighting based on the value of consumption in (7). The various approaches assuming that the distribution is optimal, or that it is described in a separate term as in decompositions (4)–(6), make it possible to focus on total consumption without even specifying further the distributive judgments. The function $\bar{W}(x_1, \dots, x_n)$ or $W(u_1(x_1), \dots, u_n(x_n))$ that is referred to in these constructions could be based on many different sorts of evaluation of individual well-being. That can be viewed as an advantage because it makes a focus on total expenditure compatible with many different distributive principles. But as soon as

one wants to also evaluate the distribution, it creates a danger. Indeed, formulae such as (5) and (6) naturally suggest to take the monetary value of individual consumption as the measure of individual well-being, and this is indeed done explicitly in (7). It is, however, important to be aware that the theory itself does not give any reason to adopt this measure of individual well-being. Independent arguments are needed. Here is an example of such an argument. In the theories of justice proposed by John Rawls (1971) or Ronald Dworkin (2000), social justice consists in allocating resources in a fair way, letting individuals make use of the resources at their disposal according to their own conception of the good life. If one ignores differences in internal resources (talent, disabilities), the fair distribution is that which maximizes the share of resources of those who have the least—the market valuation appearing convenient for the comparison of resource shares.¹² In sum, individual wealth is then a suitable measure for interpersonal comparisons. The market value of individual consumption can then be defended as a reasonable proxy for comparisons of wealth.

If personal abilities and needs are unequal, however, this line of argument no longer supports the market value of consumption as a proper metric. A practically important example concerns the valuation of leisure time, which is part of consumption broadly construed. The two variants of the revealed preference argument that have been invoked in this section both advocate using individuals' net wage rate as the proper valuation of leisure time. There is nothing objectionable with this practice if one refrains from making interpersonal comparisons. But this valuation of leisure time makes it tempting to compare

¹¹ This is because the exact equality would hold for $\alpha_i = 2r(i) - 1$ (letting $r(i)$ denote i 's rank), and for large numbers the ratio $(2r(i) - 1)/(2r(j) - 1)$ is not very different from $r(i)/r(j)$. Such rank-order weights are, however, very special and cannot be used in general. Indeed, Hammond (1978) shows that the partial ordering of allocations generated by this approach is compatible with a Paretian social welfare function $\bar{W}(x_1, \dots, x_n) = W(u_1(x_1), \dots, u_n(x_n))$ only if all individual Engel curves are linear and identical.

¹² Rawls's list of primary goods includes nonmarketed items such as basic freedoms, the powers and prerogatives of positions of responsibility, and the social bases of self-respect. Wealth is only one item.

full incomes (i.e., earnings plus the value of leisure) across individuals and to consider that those with a greater full income are better off and that it would be nice to reduce inequalities in full incomes. This is a case in which the extrapolation from total valuation to interpersonal comparisons is highly questionable.¹³ It is a particular instance of the frequent cases in which individuals face different prices on the market, raising the question of finding a proper deflator. This issue is examined in the next subsection.

Other limitations of the theory described in this section are linked to the fact that, with the exception of the decompositions (2)–(5), the purpose is only to justify the claim that if, for the prices p prevailing at X , one has $pX > pY$, then X is better than Y in some relevant sense. Beyond the obvious fact that this yields a very partial criterion, two specific issues are worth stressing.

One important shortcoming—which also plagues the decompositions (4) and (5)—is that this approach cannot be used to compare the situations of two *different* populations, as in international comparisons, or intertemporal comparisons over substantial time spans. It can only serve to examine a change of consumption for a given population with given preferences. Sen (1976) examines this problem and shows that, under the mild additional principle that only the statistical distribution of individual situations matters, not the size of the population, the approach can be extended to comparisons between populations in a limited way: specifically, one can then check if the population in one country

is better off than if it were served the *distribution* of consumption of another country. Concretely this is done by checking if $\alpha_1 px_1 + \dots + \alpha_n px_n > \alpha_1 py_1 + \dots + \alpha_n py_n$, where $i = 1, \dots, n$ refer to n -quantiles instead of concrete individuals. But this sort of comparison is still very partial—it is for instance possible to find in some cases that country A is better off (in its eyes) than country B and that country B is also better off (in its eyes) than country A .

Another limitation is that, not only is the criterion silent in some comparisons, but when it is conclusive, one does not get a *quantitative* evaluation of welfare change, as would be obtained with an index number such as GDP in volume. It is only in the decompositions (4)–(5) that one can give a quantitative meaning to the evolution of total consumption at certain well-defined prices. More direct attempts to obtain numerical estimates can be found in the theory of index numbers.

2.5 Toward an Index

Initiated by Fisher (1922), the theory of index numbers has burgeoned in three directions. One consists in defining desirable properties for the price and quantity indexes $P(p^1, p^0, x^1, x^0)$ and $Q(p^1, p^0, x^1, x^0)$ (where p^t and x^t denote the price and quantity vectors at $t = 0, 1$). Here are three examples of such properties. The first says that the product of the price index and the quantity index must equal the value index. The second says that when the price vector is unchanged, the quantity index must equal the value index. The third says that reversing time should yield inverse values of indexes.

$$(8) \quad P(p^1, p^0, x^1, x^0)Q(p^1, p^0, x^1, x^0) = \frac{p^1 x^1}{p^0 x^0};$$

$$(9) \quad Q(p, p, x^1, x^0) = \frac{px^1}{px^0};$$

$$(10) \quad P(p^1, p^0, x^1, x^0)P(p^0, p^1, x^0, x^1) = 1.$$

¹³ Equalizing full incomes across individuals would imply that those with greater productivity would have strictly smaller budget sets than those with lower productivity, in a strange reversal from the laissez-faire situation. Moreover, they would be forced to work to pay their taxes, implying what Dworkin (2000) called a “slavery of the talented.” The literature on fairness has proposed various reasonable ways of comparing the resource shares of individuals with unequal productivity (see footnote 58).

W. E. Diewert (1992b) provides a list of twenty-one axioms of this sort, and shows that the pair of Fisher indexes

$$P^F(p^1, p^0, x^1, x^0) = \left(\frac{p^1 x^0}{p^0 x^0} \frac{p^1 x^1}{p^0 x^1} \right)^{1/2},$$

$$Q^F(p^1, p^0, x^1, x^0) = \left(\frac{p^0 x^1}{p^0 x^0} \frac{p^1 x^1}{p^1 x^0} \right)^{1/2},$$

satisfies all of them and is the only one to do so (the characterization, however, involves quite controversial axioms).¹⁴ Observe for instance that the Laspeyres and Paasche indexes do not satisfy (10). The axioms in this first branch of the theory are not, however, directly connected with welfare considerations and that is a serious shortcoming.

The second branch of the theory of index numbers seeks indexes that depend only on price and quantity data but are good approximations of measures of welfare change such as equivalent variations. Diewert (1976, 1992a) suggests an approach that consists in seeking functional forms for the expenditure functions that are (1) flexible enough so that they provide good approximations to the second order of any twice differentiable expenditure function, and (2) simple enough so that the corresponding equivalent variation depends only on price and quantity data. The obtained measure of equivalent variation is called “superlative” by Diewert, who provides several examples, among which one involves the geometric

mean of the Laspeyres and Paasche indexes (i.e., the Fisher index) and another involves the arithmetic mean:¹⁵

$$(11) \quad p^0 x^0 \left[\left(\frac{p^0 x^1}{p^0 x^0} \frac{p^1 x^1}{p^1 x^0} \right)^{1/2} - 1 \right];$$

$$(12) \quad p^0 x^0 \left[\frac{1}{2} \frac{p^0 x^1}{p^0 x^0} + \frac{1}{2} \frac{p^1 x^1}{p^1 x^0} - 1 \right].$$

A limitation of this approach is that it is strictly local and cannot be used for the study of large welfare changes. Even for local applications, it is puzzling that different expressions, which equally pretend to be good approximations of the equivalent variation, may have different signs in certain cases.¹⁶ No approximation result can circumvent the hard fact that price and quantity data cannot completely determine the sign of welfare changes, even locally.

The third branch, initiated by Samuelson and S. Swamy (1974), seeks indexes which, contrary to those examined in the previous paragraphs, may depend on individual preferences and not just on prices and quantities, while retaining typical features

¹⁵ The second formula was proposed in Martin L. Weitzman (1988).

¹⁶ Even when the Laspeyres and Paasche indexes are arbitrarily close to unity, their geometric mean can be lower than unity while their arithmetic mean is greater. The explanation of this apparent contradiction in the results is that each formula for the equivalent variation is correct only when the true expenditure function is exactly equal to a particular flexible function (one for each formula) over a range spanning the initial utility u^0 and the final utility u^1 . Otherwise there is an error term, which converges to zero when u^1 tends to u^0 but may remain larger than the welfare change throughout. Diewert (1992a) provides results to the effect that the different formulae approximate each other to the second order, but only for the case when p^1 and x^1 are proportional to p^0 and x^0 , respectively—observe that x^1 being proportional to x^0 is enough for the two expressions (11) and (12) of the equivalent variation to be equal.

¹⁴ In addition to (8) and (10) and the requirement that the indexes be positive, the characterization relies on the following properties, which impose symmetry among the weights of periods 0 and 1:

$$P(p^1, p^0, x^1, x^0) = P(p^1, p^0, x^0, x^1), \\ Q(p^1, p^0, x^1, x^0) = Q(p^0, p^1, x^1, x^0).$$

of index numbers. A particularly interesting proposal is the money-metric utility function $e(p^*, u(x))$, which is derived from the expenditure function $e(p, u)$ and computes the minimal expenditure required to obtain the same satisfaction as with x at reference prices p^* —it is therefore measured in monetary units. Another famous proposal is the ray utility function defined by the property that the individual is indifferent between x and $u(x)x^*$, for a reference bundle x^* . This utility is unitless but corresponds to fractions of a bundle and can therefore be also treated like a quantity index.¹⁷ This approach is more demanding because it requires data on preferences, not just on prices and quantities. The counterpart is that, contrary to the previous approaches, these indexes do always reflect preferences. Another interesting feature is that they can serve for comparisons between different individuals with different preferences, if an ethical justification can be provided for such comparisons. Money-metric utilities (also named “equivalent incomes” by King 1983) have been popular in the 1970s and early 1980s,¹⁸ but various objections have been raised against using such utility functions in the context of social welfare evaluation. These objections will be examined in Section 4.

On the basis of Pollak’s (1981) social expenditure function, it has been proposed to apply the money-metric utility function at the social level directly. The function $V(x_1, \dots, x_n)$ defined in (3) is indeed a social money-metric utility function, with

reference prices p^* . Being expressed in monetary units, its evolution in time can be compared to that of GDP, no matter what social welfare function $W(u_1(x_1), \dots, u_n(x_n))$ is adopted. Along this vein, Jorgenson (1990, 1997) and Slesnick (2001), for instance, boldly adopt a particular specification for the indirect utility function of households, estimate it on demand data (making restrictions in order to be able to perform the estimation on aggregate demand), define a social welfare function that incorporates some degree of inequality aversion, and compute an index of social welfare with $V(x_1, \dots, x_n)$. The specific calibration of utility functions chosen by Jorgenson and Slesnick is not provided much justification, apart from the aesthetic feature that these functions essentially correspond to the logarithm of expenditure. The main limitation of the Pollak approach, obviously, is that it needs a well-defined social welfare function as an input and does not help at all in its construction. This approach is useful only in the translation of a social welfare function into a monetary index. Moreover, it cannot in general be extended to comparisons of situations across populations with different preferences. Such an extension would require that for all profiles of utility functions $(u_1, \dots, u_n)$, $(u'_1, \dots, u'_n)$, with associated social welfare functions W, W' , the corresponding V, V' functions should be such that

$$(13) \quad V(x_1, \dots, x_n) = V'(x'_1, \dots, x'_n)$$

if and only if

$$W(u_1(x_1), \dots, u_n(x_n)) = W'(u'_1(x'_1), \dots, u'_n(x'_n)).$$

This is a very demanding property. It can be satisfied when the original social ordering is based on money-metric utilities at the individual level (for the same reference price p^* as that used for V, V'), but it is hard to see how its

¹⁷This function, discussed, among others, in Sten Malmquist (1953), Samuelson (1977), Elisha Pazner (1979), is also linked to the “distance function” studied in Angus Deaton (1979), which can be defined by the formula $d(v, q) = 1/u(x)$, where u is the ray utility function with reference q and v is the utility level (for any given utility function) of the individual at x .

¹⁸See in particular Deaton and John Muellbauer (1980).

scope of validity could be broader.¹⁹ Therefore, this approach does not make much progress compared to the application of money-metric utilities at the individual level.

In conclusion, although the market value of total consumption at initial or final prices can be given some justification as a partial indicator (partial in at least one of two ways: an incomplete ordering; ignoring distribution) of welfare variation, it is of little use for comprehensive and quantitative assessments. Particularly limiting is the restriction in most approaches described here to the comparison of bundles or budget sets for *given* preferences, rather than across different situations involving *different* preferences.²⁰ Money-metric utilities at the individual level appear more promising in this respect and are the topic of section 4.

3. NNP and Green Accounting

Total consumption should ideally be computed over a lifetime, or at least over some reasonable life horizon. Statistics like

¹⁹ Consider a social welfare function $W(u_1(x_1), \dots, u_n(x_n))$. It can be rewritten as a function of individual money-metric utilities:

$W(u_1(x_1), \dots, u_n(x_n)) = \hat{W}(e(p^*, u_1(x_1)), \dots, e(p^*, u_n(x_n)))$. The function $V(x_1, \dots, x_n)$ is simply the value of the expenditure function of $\hat{W}$ at price $(1, \dots, 1)$:

$V(x_1, \dots, x_n) = \min e_1 + \dots + e_n \text{ s.t. } \hat{W}(e_1, \dots, e_n) \geq W(u_1(x_1), \dots, u_n(x_n))$.

Therefore, (13) means that the corresponding $\hat{W}$ and $\hat{W}'$ have the same expenditure function at price $(1, \dots, 1)$. This, in itself, does not put any logical constraint on the ordinal properties of $\hat{W}$ and $\hat{W}'$. But the identity of expenditure functions at a particular price seems an unlikely coincidence. Now, if $\hat{W}$ and $\hat{W}'$ have the same expenditure function at all prices, they are identical, which implies that the social ordering is always defined as the same social welfare function $\hat{W}$ on individual money-metric utilities.

²⁰ The following view, quoted from Weitzman (2001, p. 8) justifying an assumption of identical preferences over two countries, is probably shared by many authors: "Unless preferences are postulated to be comparable in some way across any two situations, it is impossible to make rigorous general welfare comparisons." It will be argued in section 4 that this is unduly pessimistic.

national income or GDP add up consumption and savings (or investment). One can of course take savings as an argument of individual utility and proceed as in the previous section, but that is much less satisfactory than a rigorous treatment deriving the utility of savings from the prospect of future consumption (or bequest). Moreover, it seems important to worry about the sustainability of current living standards and ethically appealing to incorporate future generations into the computation of social welfare. In this perspective, the depletion of natural resources should not just be counted as immediate consumption if it reduces the prospects for future consumption even in the very long run. Therefore, the idea of incorporating the depreciation of the stock of capital, as NNP²¹ does for invested capital, appears a reasonable move in this perspective. In this section we examine the pros and cons of computing such aggregates as NNP or "genuine savings,"²² which have recently become popular.

3.1 Wealth, Social Income, Welfare

NNP has long been understood as connected to the idea of sustainability. Sustainable consumption is defined in Nordhaus and Tobin (1973) as "the amount society can consume without shortchanging the future" (p. 38). More precisely, it can be defined either as the total amount of consumption compatible with keeping per capita consumption constant in the future

²¹ Reference to NNP is made here because this is the usual acronym used in the literature on green accounting. The relevant difference with GDP is the difference between "gross" and "net." The other difference (between "domestic" and "national") is important practically (many observers regret that national accounts switched from GNP to GDP at a time when globalization made domestic production less relevant than national income) but does not raise any complex problem and has nothing to do with the topic of this section.

²² The expression "genuine savings" is due to Kirk Hamilton and Michael Clemens (1999).

(implying a reduction in capital per capita compensated by technical progress) or as the (lower) amount that is compatible with keeping a growth rate of per capita consumption equal to the rate of technical progress.²³ In absence of population growth and technical progress, sustainable consumption is the amount of consumption that is compatible with keeping the stock of capital constant (ignoring possible complications with the valuation of multiple capital goods), which corresponds to NNP as noted by Nordhaus and Tobin.

A more recent line of research has sought to relate NNP to intergenerational social welfare. Starting from the point of view of the previous section, the most natural idea in order to capture intertemporal possibilities is to extend the measure of total consumption by incorporating the whole future stream of consumptions so as to obtain a “wealth-like” quantity (Samuelson 1961). As NNP can be viewed as a measure of the income generated by the available stock of capital, one can suspect a close link between wealth, measured as the present discounted value of consumption, and NNP. Such link was first established by Weitzman (1976), and I will now discuss this approach. Much of the literature on green accounting²⁴ focuses on optimal growth paths, which is of little practical interest. Following Partha S. Dasgupta and Karl-Goran Mäler (2000) and Asheim (2007a), one can however develop a reasoning for any arbitrary path.

²³ Nordhaus and Tobin (1973, p. 34) favor the second approach with the following argument: “Our proposed measure . . . is more austere and, we believe, more consonant with revealed social preference. We do not observe current generations consuming capital on the grounds that successors will reap the benefits of technological progress.” Taking account of natural capital, one might object that the frivolous approach, or even worse, has actually prevailed.

²⁴ Syntheses of the theory are provided by Geir B. Asheim (2000) and Geoffrey M. Heal and Bengt Kriström (2005).

Suppose that consumption C_t (a real number) over the time interval $[0, +\infty)$ is limited by the evolution of a real vector of stocks denoted S_t , which includes produced capital, natural resources, technological knowledge, and any other kind of social and human capital that influences consumption possibilities. The variation of these stocks over time is governed by a system of equations

$$\frac{dS_t}{dt} = F(S_t, C_t, R_t).$$

In these equations, R_t is a vector of instruments such as investment in research or replenishment and F is a vector of functions (one for each component of S_t). There is no exogenous technical progress, which is partly justified by the fact that S_t can be as comprehensive as one wishes.²⁵ The population is assumed to remain constant.²⁶

Consider an arbitrary program²⁷ that defines the evolution path of all the variables C, S, R over any subinterval $[t, +\infty)$ as a function of the stock S_t available in t . I assume throughout this section that this program is independent of t and time-consistent. This program makes it possible to define a function computing the present value of the future consumption path

$$(14) \quad Z(S_t) = \int_t^{+\infty} e^{-\delta(\tau-t)} C_\tau(S_t) d\tau.$$

The corresponding constant equivalent consumption (i.e., the constant amount of consumption yielding the same present

²⁵ Climate change and variations in terms of trade, however, should require making the technology time dependent.

²⁶ On the variable population case, see Kenneth J. Arrow, Dasgupta, and Mäler (2003b).

²⁷ This is called a “resource allocation mechanism” in the literature. In Dasgupta and Mäler (2000), a “program” is a particular path of all variables over the future.

discounted value) equals $Z(S_t)/\int_t^{+\infty} e^{-\delta(\tau-t)} d\tau = \delta Z(S_t)$.

Recall that this program need not be optimal or efficient. One can nonetheless compute the corresponding “accounting prices” $p_t = \nabla Z(S_t)$, where ∇ denotes the gradient.²⁸ By the assumption that the program is independent of t and time-consistent, the evolution of $Z(S_t)$ over time equals

$$\frac{d}{dt} Z(S_t) = p_t \frac{dS_t}{dt}.$$

From (14) one also directly computes

$$\begin{aligned} \frac{d}{dt} Z(S_t) &= \delta Z(S_t) - C_t(S_t) \\ &\quad + \int_t^{+\infty} e^{-\delta(\tau-t)} \nabla C_\tau(S_t) \frac{dS_t}{dt} d\tau, \end{aligned}$$

in which the last term is null because time consistency requires $C_\tau(S_t) = C_\tau(S_{t+dt})$ along the path.²⁹

Combining these two equalities, one obtains

$$(15) \quad \delta Z(S_t) = C_t(S_t) + p_t \frac{dS_t}{dt}.$$

On the left-hand side, one has the constant equivalent consumption. On the right-hand side, the expression looks very much like NNP. What is remarkable about this formula is that, while $Z(S_t)$ depends on the whole future of the economy, the expression on the right-hand side depends only on current variables of flows and prices. It therefore seems to

invalidate Samuelson’s (1961) conclusion that in a dynamic setting welfare cannot be measured in terms of income but only in terms of wealth, with serious difficulties because “there is so much ‘futuraity’ in any welfare evaluation of any dynamic situation as to make it exceedingly difficult for the statistician to approximate to the proper wealth comparisons” (p. 53).

3.2 Is that NNP?

Before one can deduce from this reasoning that NNP is a good measure of social welfare, many obstacles have to be taken into consideration. They fall under two headings. First, (15) is not the usual notion of NNP from national accounts. Second, (15) need not be a good measure of social welfare, even assuming a representative agent or an altruistic dynasty, as is typically done in this literature.

The formula in (15) does not correspond to standard NNP and is hard to estimate in practice because the stock S_t encompasses much more than physical capital and because accounting prices are not market prices unless, as assumed in Weitzman’s results, there is a complete set of markets and the path is a market equilibrium in which an infinitely lived representative agent maximizes a utility function that is ordinally equivalent to the discounted sum defining $Z(S_t)$. When this is not the case, in particular when the market equilibrium is not efficient because, for instance, the depletion of natural resources involves a tragedy of the commons between different agents or because the environment involves external effects, the market prices are not reliable indicators for the estimation of accounting prices.³⁰ Valuing natural resources may be rather difficult, and valuing other stock variables such as technological knowledge or social capital may be even harder. It already appears

²⁸ The expression $\nabla Z(S_t)$ is the vector with components $\partial Z / \partial S_{tk}$, for every stock variable k .

²⁹ Rigorously, one should write functions in which the value of initial stocks is distinguished from its time: $C_\tau(S; t)$, $S_\tau(S; t)$, $R_\tau(S; t)$. Time consistency requires: $C_\tau(S; t) = C_\tau(S_\tau(S; t); t')$ for all $\tau \geq t' \geq t$. Time invariance requires: $C_{t+\tau}(S; t) = C_{t+\tau}(S; t')$ for all $\tau \geq 0$ and all t, t' .

³⁰ See, e.g., Dasgupta and Heal (1979). More recent syntheses include A. Myrick Freeman (1993), Arrow, Dasgupta, and Mäler (2003a), Heal and Kriström (2005).

very demanding that S_t should not only incorporate natural resources but also any other stock variable that influences future consumption prospects. But measuring quantities is the easiest part of the exercise. It is the evaluation of accounting prices p_t that bears the whole load of “futurity” that Samuelson found daunting and that has not magically disappeared, unfortunately, in the derivation of (15).

3.3 Is That Social Welfare?

The second set of issues has to do with the point that the connection between (15) and social welfare is problematic. Many authors have criticized Weitzman (1976, 1998) for assuming that social welfare is the discounted sum of consumption, which implies in particular the absence of inequality aversion across generations. Moreover, when consumption is multidimensional the level of NNP cannot be so easily interpreted and a variant of Weitzman’s holds only for infinitesimal variations of NNP through the proportionality between marginal utility and market prices for commodities. I examine these issues by modifying the framework slightly.

Assume now that C_t is a vector (including also labor, in particular) and that social welfare is computed as a discounted sum of utility

$$Z^*(S_t) = \int_t^{+\infty} e^{-\delta(\tau-t)} U(C_\tau(S_t)) d\tau,$$

which takes account of inequality aversion via decreasing marginal utility. From similar computations as in the previous subsection, the constant-equivalent flow of utility equals:

$$(16) \quad \delta Z^*(S_t) = U(C_t(S_t)) + p_t \frac{dS_t}{dt},$$

where the accounting prices are computed anew with respect to $Z^*(\cdot)$. We also have

$$\frac{d}{dt} Z^*(S_t) = p_t \frac{dS_t}{dt}.$$

Moreover, differentiating (16) and letting $q_\tau = \nabla U(C_\tau(S_t))$, one computes:

$$\delta \frac{d}{dt} Z^*(S_t) = q_t \frac{d}{dt} C_t(S_t) + p_t \frac{d^2 S_t}{dt^2} + \frac{dp_t}{dt} \frac{dS_t}{dt}.$$

Suppose that NNP_t is computed with prices $(\hat{q}_t, \hat{p}_t)$ proportional to (q_t, p_t) and is therefore equal to $\alpha_t(q_t C_t + p_t(dS_t/dt))$ for some $\alpha_t > 0$. Its variation at fixed prices is then proportional to

$$q_t \frac{dC_t}{dt} + p_t \frac{d^2 S_t}{dt^2}$$

and is, therefore, *not* proportional to the variation in constant-equivalent utility, for lack of a term involving the change in accounting prices of S_t . Another option, proposed by Asheim and Weitzman (2001), is to deflate $(\hat{q}_t, \hat{p}_t)$ by a *consumer* (not an NNP) price index π_t satisfying the Divisia property

$$(17) \quad \frac{1}{\pi_t} \frac{d\pi_t}{dt} = \frac{\frac{d\hat{q}_t}{dt} C_t}{\hat{q}_t C_t}.$$

One then obtains

$$\begin{aligned} \frac{d}{dt} \left(\frac{NNP_t}{\pi_t} \right) &= \frac{\hat{q}_t}{\pi_t} \frac{dC_t}{dt} + \frac{d(\hat{q}_t/\pi_t)}{dt} C_t \\ &\quad + \frac{\hat{p}_t}{\pi_t} \frac{d^2 S_t}{dt^2} + \frac{d(\hat{p}_t/\pi_t)}{dt} \frac{dS_t}{dt}, \end{aligned}$$

where the second term vanishes by (17). This can be rewritten

$$\begin{aligned} \frac{d}{dt} \left(\frac{NNP_t}{\pi_t} \right) &= \frac{1}{\pi_t} \left(\hat{q}_t \frac{dC_t}{dt} + \hat{p}_t \frac{d^2 S_t}{dt^2} + \alpha_t \frac{dp_t}{dt} \frac{dS_t}{dt} \right) \\ &\quad + \frac{1}{\pi_t} \frac{d\alpha_t}{dt} p_t \frac{dS_t}{dt} - \frac{1}{\pi_t^2} \frac{d\pi_t}{dt} \hat{p}_t \frac{dS_t}{dt} \\ &= \frac{\alpha_t}{\pi_t} \left(\delta - \frac{1}{\pi_t} \frac{d\pi_t}{dt} + \frac{1}{\alpha_t} \frac{d\alpha_t}{dt} \right) \frac{d}{dt} Z^*(S_t), \end{aligned}$$

which shows that with this price index, the variation of real NNP_t is proportional to the variation of $Z^*(S_t)$ provided the “real interest rate” $\delta - (1/\pi_t)(d\pi_t/dt) + (1/\alpha_t)(d\alpha_t/dt)$ is positive.³¹

Building on Samuelson (1961), Heal and Kilström (2005, 2008) propose to seek a more direct adaptation to the dynamic context of the local version of the revealed preference argument of the previous section. Consider the value of consumption at supporting prices $q_\tau = \nabla U(C_\tau(S_t))$.³²

$$\bar{Z}^*(S_t) = \int_t^{+\infty} e^{-\delta(\tau-t)} q_\tau C_\tau(S_t) d\tau.$$

One computes, for any particular k th component of S_t :

$$\begin{aligned} \frac{\partial}{\partial S_{tk}} \bar{Z}^*(S_t) &= \int_t^{+\infty} e^{-\delta(\tau-t)} \nabla U(C_\tau(S_t)) \frac{\partial}{\partial S_{tk}} C_\tau(S_t) d\tau \\ &= \frac{\partial}{\partial S_{tk}} Z^*(S_t). \end{aligned}$$

As a consequence,

$$\frac{d}{dt} \bar{Z}^*(S_t) = \frac{d}{dt} Z^*(S_t) = p_t \frac{dS_t}{dt}.$$

The evolution of welfare can, therefore, also be interpreted in terms of a notion of intertemporal income, computed over the whole consumption path at supporting

future prices, and measurable by the evolution of stocks at current accounting prices. As before, one can also directly compute

$$\frac{d}{dt} \bar{Z}^*(S_t) = \delta \bar{Z}^*(S_t) - q_t C_t(S_t),$$

implying that

$$\delta \bar{Z}^*(S_t) = q_t C_t(S_t) + p_t \frac{dS_t}{dt}.$$

If the proportionality coefficient α_t that relates $\hat{q}_t$ to q_t is constant over time, one then obtains an interpretation of NNP in terms of intertemporal income evaluated at prices $\hat{q}_\tau$, as a direct extension of (15):

$$\begin{aligned} NNP_t &= \alpha_t \left(q_t C_t(S_t) + p_t \frac{dS_t}{dt} \right) \\ &= \delta \int_t^{+\infty} e^{-\delta(\tau-t)} \hat{q}_\tau C_\tau(S_t) d\tau. \end{aligned}$$

The main achievement of this branch of the literature has been to show that the assessment of the evolution of a present discounted value (intertemporal social welfare) can be made with instantaneous price and quantity variables, namely, the variation of stocks at current accounting prices, or “genuine savings,” $p_t dS_t/dt$. However, the limitations of the revealed preference approach listed in the previous section apply equally well to these extensions to the dynamic context, with the additional twist that the prices p_t are even harder to estimate. In a similar fashion as in the previous section, one cannot connect NNP or intertemporal income to a meaningful *level* of welfare, but only to evolutions of welfare over time or in response to policy changes. Comparisons across countries are still problematic, not only when the populations have different preferences but also when the countries follow different programs because their respective accounting prices are then different.

³¹ Note that $\frac{1}{\pi_t} \frac{d\pi_t}{dt} - \frac{1}{\alpha_t} \frac{d\alpha_t}{dt} = \frac{\frac{dq_t}{dt} C_t}{q_t C_t}$.

³² Heal and Kilström (2008) assume a perfect market with an infinitely-lived representative agent. Their main result can be adapted to our framework in which the growth path is governed by a possibly imperfect “program.”

As explained in the beginning of this section, NNP is also commonly viewed as a measure that is less myopic than GNP and, putting intergenerational welfare aside, provides indications of sustainability. In this respect one should not forget an obvious limitation of a measure of wealth or sustainable consumption. Namely, it describes the opportunities of the present time but tells little about the real prospects for future generations if one does not assume that a socially optimal growth path is maintained.³³ Indeed, depending on how the present generation manages its stock of wealth, the future may look very different. A greater level of wealth or sustainable consumption is compatible with lower sustainability if the present generation consumes more. I will come back to this issue at the end of this section.

3.4 *To Discount or Not to Discount?*

Another critical issue is whether social welfare, in the intergenerational context, should be defined as a discounted sum of utility.³⁴ Discounting appears to violate the most minimal requirement of impartiality between individuals living at different times and many authors, after Henry Sidgwick (1907) and F. P. Ramsey (1928), have rejected it vehemently.³⁵ In absence of discounting, however, a social welfare function with finite

aversion to inequality faces a problem of infinite sum. A vast literature has addressed this issue, with many recent developments.³⁶ A striking result, due to William R. Zame (2007), is that no constructible³⁷ social ordering over infinite utility streams satisfies Weak Pareto (i.e., if utility at each period is increased, this is a strict improvement) and finite anonymity (i.e., permuting the utility of two periods gives an equivalent stream). In order to get a sense of the problem, consider two slightly stronger requirements, Strong Pareto (i.e., if utility is increased at one period and decreased at none, this is a strict improvement) and adjacent anonymity (i.e., permuting the utilities of adjacent periods gives an equivalent stream, even when infinitely many pairs are permuted).³⁸ The latter requirement means in particular that $(0, 1, 0, 1, \dots)$ is equivalent to $(1, 0, 1, 0, \dots)$, and that, starting the transpositions from the second period, $(1, 0, 1, 0, 1, \dots)$ is equivalent to $(1, 1, 0, 1, 0, \dots)$. By transitivity, one concludes that $(0, 1, 0, 1, \dots)$ is equivalent to $(1, 1, 0, 1, 0, \dots)$, a clear violation of Strong Pareto.³⁹ It is clear that such difficulties, in this particular field, come from the infinite horizon. These problems are intellectually

³⁶ A sample of such developments, with ample reference to the recent and less recent literature, can be found in John E. Roemer and Kotaro Suzumura (2007).

³⁷ A constructible object can be explicitly defined, whereas a nonconstructible one can only be proved to exist (with arguments like the axiom of choice). For practical purposes of social evaluation, the absence of a constructible ordering is like a nonexistence result. See also L. Lauwers (2006) for a very similar result.

³⁸ Restricting adjacent anonymity to a finite number of permutations gives a condition that is equivalent to finite anonymity. To see the difference between finite anonymity and adjacent anonymity, observe for instance that the overtaking criterion (which weakly prefers a utility stream if and only if the sum of utilities is at least as great for all dates except a finite number) violates adjacent anonymity but satisfies finite anonymity.

³⁹ Discounted utilitarianism is transitive, complete and constructible, it satisfies Strong Pareto but violates adjacent (and finite) anonymity. The liminf criterion satisfies adjacent anonymity and is transitive, complete and constructible, but it violates Strong (and Weak) Pareto.

³³ The problem is formally similar to one that was already mentioned in the previous section in the static context: total consumption does not guarantee greater social welfare unless the distribution is optimal.

³⁴ See Asheim (2007a) for an extension of some of the above results to a more general objective than discounted utilitarianism.

³⁵ It is important to distinguish the issue of the discount rate as a pure expression of time preference in the computation of intergenerational welfare, which is the topic of this subsection, from the more applied issue of finding a good discount rate to be used in the computations of the present discounted value of income streams for investment projects. Even with a quasi-zero time preference for utility, the discount rate for money may be substantial if future income is greater than present income and marginal utility is decreasing (due for instance to inequality aversion across generations).

fascinating and challenging, but for practical purposes, one may be worried about making concrete decision criteria depend in a deep fashion on the nonfiniteness of the horizon.

A more promising—and realistic—approach might be to work with a finite but uncertain horizon. Interestingly, this approach may not bring back a standard discounted sum of utilities with a discount factor equal to the survival probability. The literature addressing the problem in this way is sparse⁴⁰ and this is probably a promising field of research, albeit full of difficulties because it combines the theory of social choice under uncertainty with the study of population ethics.⁴¹ Indeed, in order to assess a prospect with uncertain horizon, one needs a well defined measure of intertemporal social welfare for any finite population size. For instance, take Yew-Kwang Ng's (1986) number-dependent utilitarian criterion

$$W(u_1, \dots, u_n) = \frac{f(n)}{n} \sum_{i=1}^n u_i,$$

where f is a nondecreasing function such that $1 \leq f(n) \leq n$ for all n . This criterion corresponds to classical utilitarianism (generally considered to be excessively populationist) for $f(n) = n$ and to average utilitarianism (considered Malthusian) for $f(n) = 1$. The situation that is the most favorable to obtaining a form of discounting is when there is a fixed infinite sequence of utilities $(u_1, u_2, \dots)$ and the risky scenario says that this sequence may stop at any time, the conditional probability of surviving one more period given that one has reached period t being equal

to a constant $\pi < 1$. The probability of the world ending just after period t is $\pi^t(1 - \pi)$. To simplify, we suppose that there is only one individual per generation.

It is reasonable to compute social welfare for a risky scenario as the expected value of social welfare over each possible state of the world. The expected social value of the whole scenario is then

$$\sum_{t=1}^{\infty} \pi^t (1 - \pi) \frac{f(t)}{t} \sum_{i=1}^t u_i.$$

The weight of u_t in this expression is proportional to

$$\sum_{t \geq T} \pi^t \frac{f(t)}{t}.$$

This yields the awaited discount factor π if $f(t) = t$ (classical utilitarianism), but this series decreases at a greater pace, implying a greater preference for the present, if $f(t)/t$ is decreasing. Intuitively, if large populations do not bring much social value by their size, it becomes relatively more important to enhance the well-being of earlier generations which are more likely to exist.⁴²

A more modest option, which may be safer to explore for empirical applications, consists in abandoning the project of a global intergenerational welfare criterion and in focusing on two simpler issues. One issue is the measurement of social welfare for certain generations or cohorts, for which an easy extension of the static analysis can be made. The other issue is forecasting the evolution of social welfare for the future generations.⁴³ For the latter purpose, a

⁴⁰ Dasgupta and Heal (1979) is a classical reference. A recent paper addressing this problem, and showing that the discount factor need not correspond to the probability of survival, is Antoine Bommier and Stéphane Zuber (2008).

⁴¹ For a synthesis on population ethics, see Charles Blackorby, Walter Bossert, and David Donaldson (2005). They do not, however, specifically study the link between uncertainty and discounting.

⁴² Introducing inequality aversion in the criterion would also affect the discount rate.

⁴³ This two-tier approach is defended, e.g., by Eric Neumayer (2004), who points out that it is illusory to seek to assess social welfare and sustainability in a single measure.

comparison between current consumption and sustainable consumption is a natural idea, and the approach presented in the previous subsection may be helpful in a different way. Indeed, one can rewrite (16) as

$$\delta Z^*(S_t) - U(C_t) = p_t \frac{dS_t}{dt}.$$

This expression is informative about the future evolution of instantaneous welfare $U(C_t)$ for the following reason. Along the program under consideration, utility has to be lower than $U(C_t)$ at some future time if $\delta Z^*(S_t) - U(C_t) < 0$, because $\delta Z^*(S_t)$ is a weighted average of $U(C_\tau)$ at all future dates. Sustainability understood as a non-decreasing utility path over the whole future therefore requires non-negative genuine savings, i.e., $p_t dS_t/dt \geq 0$.⁴⁴ Importantly, this reasoning does not ask us to accept the discounted sum $Z^*(S_t)$ as a criterion of social welfare because it simply uses $\delta Z^*(S_t)$ as a summary of the future evolution of $U(C_\tau)$. Moreover, this necessary condition of sustainability is valid for any arbitrary value of the discount rate δ , and therefore the requirement $p_t dS_t/dt \geq 0$ must be satisfied for the prices p_t corresponding to all possible values of δ —recall that p_t is the gradient of $Z^*(S_t)$ and thereby depends on δ . For a high value of δ , p_t depends mostly on the influence of S_t on $U(C_\tau)$ for τ close to t . For δ close to zero, p_t depends mostly on the influence of S_t on the utility of distant generations.

This sustainability criterion, however, has two important weaknesses. First, it only gives a necessary but not sufficient condition, and

can be satisfied in cases in which the most likely scenario is a decline of utility for some period in the future. Second, the price vector p_t , being the gradient of $Z^*(S_t)$, is based on the definition of a “program,” i.e., a function that defines the impact on the future growth path of all possible counterfactual changes in the current stock S_t . It is unlikely that such a program exists or can be elaborated in practice, even for small changes in S_t in the vicinity of current values.⁴⁵ Market prices can provide the needed price parameters only if the equilibrium maximizes $Z^*(S_t)$, which is quite unlikely as already explained. Therefore, even though this approach appears to provide a convenient framework to think about sustainability and has inspired interesting applications, it is not clear whether it provides more practical help in forecasting the evolution of welfare and checking sustainability than a direct attempt at forecasting the evolution of $U(C_t)$ with a dynamic model.

4. Interpersonal Comparisons and the Equivalence Approach

The previous section addressed intertemporal issues. Another set of issues raised in section 2 has been postponed to the present section, namely, the measurement of the contribution of the *distribution* of individual consumption (or well-being) to social welfare. This requires finding a metric of individual well-being and addressing the vexed issue of interpersonal comparisons. As we have seen, the theory underlying measures of total

⁴⁴ An implication of this line of reasoning is a version of John M. Hartwick's (1977) rule, according to which utility is kept constant if genuine savings are always null. See Asheim, Wolfgang Buchholz and Cees Withagen. (2003). For syntheses on sustainability, see, e.g., Thomas Aronsson, Per-Olov Johansson, and Karl-Gustaf Lofgren (1997), Asheim (2007b).

⁴⁵ Obviously, in a determinist world there exists a function that defines the discounted sum of utilities as a function of the initial state of the world. But this function also depicts all political choices that will be made, so that if we could estimate it we would no longer need criteria to help policy decisions. The program $Z^*(S_t)$ is not such a function but is conceived as a particular policy plan which can be changed. Such a program would need to specify, e.g., how we would proceed if oil reserves were larger than expected.

consumption or the social expenditure function involves social welfare functions $\bar{W}(x_1, \dots, x_n)$ or $W(u_1(x_1), \dots, u_n(x_n))$ and has been somewhat embarrassed about defining a precise metric for interpersonal comparisons and for the calibration of inequality aversion. Such embarrassment has been compounded by the development of social choice theory under the shadow of Arrow's impossibility theorem. "This starkly negative finding became a major stumbling block to the empirical implementation of an explicit social welfare function." (Slesnick 1998, p. 2139). However, various branches of welfare economics have developed constructive proposals around the money-metric utility and similar notions. I review such proposals and explain how they circumvent the famous "stumbling block."

4.1 Arrow's Theorem

It is worth briefly recalling the main axioms of Arrow's impossibility because they provide a nice way to classify the various approaches that constitute welfare economics. The axioms listed below are not Arrow's original axioms but a convenient reconstruction that has become popular after Sen's (1970) seminal analysis. Consider the problem of ranking alternatives from a set A on the basis of the profile of individual utility functions $(u_1, \dots, u_n)$ in the population. The arguments of individual utility functions are the alternatives from A . The social ordering that ranks the alternatives is a complete and transitive binary relation R (the expression aRa' will mean that a is at least as good as a' , and aPa' will mean that a is better than a'). One such ordering must be defined for every possible profile of utility functions in a suitable domain D , and the problem is therefore to find a "social ordering functional" f defining a particular social ordering $R = f(u_1, \dots, u_n)$ for each profile in the domain. The axioms of interest for our purposes are the following:

Ordinalism: Only the ordinal preferences of the individuals should be taken into account, i.e., for all $(u_1, \dots, u_n), (u'_1, \dots, u'_n) \in D$, if they are ordinally equivalent then $f(u_1, \dots, u_n) = f(u'_1, \dots, u'_n)$.

Independence: The ranking of two alternatives depends only on the vector of individual utility levels at each of these alternatives, i.e., for all $a, a' \in A$, all $(u_1, \dots, u_n), (u'_1, \dots, u'_n) \in D$, if $(u_1(a), \dots, u_n(a)) = (u'_1(a), \dots, u'_n(a))$ and $(u_1(a'), \dots, u_n(a')) = (u'_1(a'), \dots, u'_n(a'))$, then $aRa' \Leftrightarrow aR'a'$, where $R = f(u_1, \dots, u_n)$, $R' = f(u'_1, \dots, u'_n)$.

Weak Pareto: At any given profile, if an alternative gives a greater utility to each individual than another, it is ranked higher by the social ordering, i.e., for all $a, a' \in A$, all $(u_1, \dots, u_n) \in D$, if for all i , $u_i(a) > u_i(a')$, then aPa' , where P is the asymmetric part of $R = f(u_1, \dots, u_n)$.

Arrow's theorem says that, if the domain of profiles of utility functions is rich enough and if there are more than two alternatives, these axioms imply that the social ordering must follow the strict preferences of a fixed individual ("dictator"). This is correctly interpreted as an impossibility because this kind of "dictatorship" is unpalatable. But the strength of an impossibility theorem depends on how compelling its axioms are. In the case of Arrow's theorem, the Ordinalism and Independence axioms have been found questionable by various approaches.

4.2 Escapes

Later developments of social choice theory have rejected Ordinalism, observing that when information about utility numbers can be used, the impossibility disappears.⁴⁶ Social welfare functions

⁴⁶ For a synthesis on this approach, see Sen (1999).

$W(u_1(x_1), \dots, u_n(x_n))$, where W is an increasing function that remains the same for all profiles of utility functions, yield social ordering functionals that satisfy Independence and Weak Pareto. The motivation for dropping Ordinalism, as it is often presented in the literature, is that it should be possible to compare individuals, distinguishing the better off from the worse off. Rigorously speaking, however, Ordinalism does not exclude interpersonal comparisons because it is perfectly compatible with comparing individuals in terms of wealth, for instance. It only excludes interpersonal comparisons *of utility*. One must drop Ordinalism if one considers that individual ordinal noncomparable preferences do not constitute a sufficient informational basis for interpersonal comparisons. In particular, a welfarist approach must definitely reject this axiom. But nonwelfarist approaches may also want to reject it, for instance if, in the framework at hand, the utility functions contain relevant information about the unequal talents and needs of the individuals.

For applications, it remains to determine how to measure utility. Many different approaches can (although not all of them need to) be described as differing on the measurement of utility while being similar in the fact that they apply a certain social welfare function W to the vectors of utility levels. A welfarist theory will measure utility in terms of subjective happiness or satisfaction. Nonwelfarist theories can be described as measuring utility in terms of resources, opportunities, capabilities, and so on.⁴⁷

Other branches of welfare economics have instead avoided an impossibility by rejecting or ignoring Independence while retaining

Ordinalism. A first branch was developed by Abram Bergson (1938, 1966) and Samuelson (1947, 1977), who have argued that the social welfare function $\bar{W}(x_1, \dots, x_n)$ must work with individual indifference sets (an indifference set is the generalization of an indifference curve to a space of any dimension). Samuelson (1977) takes the example of the ray utility function as a possible calibration of indifference curves permitting interpersonal comparisons based solely on ordinal noncomparable preferences. A controversy has opposed Samuelson to social choice theorists over the possibility of a Bergson-Samuelson social welfare function without interpersonal comparisons of utility.⁴⁸ While one may consider that, in some trivial sense, the ray utility function, for instance, is a utility function, it is clear that Bergson and Samuelson were right that their construction relies only on ordinal noncomparable preferences. In their approach, the expression $W(u_1(x_1), \dots, u_n(x_n))$ is just a representation in which $u_1, \dots, u_n$ can be replaced by any other representations of the same ordinal preferences, if W is correspondingly adjusted so as to keep the social ordering of allocations unchanged. What Bergson and Samuelson can be criticized for is never to have proposed a method, similar to the axiomatics of social choice, making it possible to think rigorously about the particulars of their social welfare function—how to compare indifference sets, how much priority for the worse off. In the absence of precise examples backed up with solid ethical arguments, it has been all too easy for the profession to ignore the potential of their approach.

A second branch of welfare economics in which Independence has never been considered is cost-benefit analysis. In order to apply the Kaldor-Hicks compensation tests to two alternatives a, a' , one must be able to determine a third alternative a'' that is obtained

⁴⁷ Recall that Jorgenson and Slesnick develop their measure of social welfare simply by picking a particular utility function.

⁴⁸ A review of the controversy is made by Marc Fleurbaey and Philippe Mongin (2005).

from, say, a by transfers between individuals, and is then compared to a' by the Pareto principle. This characteristically violates Independence because it requires an estimate of individual preferences over a'' in order to rank a and a' . Similarly, computations of equivalent variations or compensating variations require evaluating the minimal expenditure needed to obtain the same satisfaction as in, say, a at the prices prevailing in a' . Cost-benefit analysis based on compensation tests and on equivalent or compensating variations has been under the fire of severe criticism. This is discussed in the next subsection.

A third branch in which Independence has been ignored is the theory of fair allocation which is known in particular for the concepts of no-envy and egalitarian-equivalence.⁴⁹ The former concept requires checking that no individual would rather consume another's bundle. The latter relies on the ray utility function in the simple model of distribution of unproduced divisible commodities: an allocation is egalitarian-equivalent if it is efficient and if every individual is indifferent between his bundle at this allocation and a fraction of the reference bundle, this fraction being the same for all individuals. It has often been argued that the theory of fair allocation avoids the negative consequences of Arrow's theorem because it only seeks to select a subset of efficient allocations instead of ranking all allocations. In fact, the selection of a subset does define an ordering, albeit a coarse one. The true reason for the positive results obtained in the theory of fair allocation is that it is not loaded with Independence. Recent developments in the theory of fair allocation have produced fine-grained rankings based on axiomatic characterizations that mimic Arrow's list of axioms except that Independence is dropped or weakened and that additional fairness axioms help define

how to compare individual situations—an example is provided in the next subsection.⁵⁰ One can consider that this branch of the theory of fair allocation gives flesh to the concept of Bergson–Samuelson social welfare function by determining precisely how to compare individual indifference sets and how much priority to give the worse off.

4.3 Equivalent Sets, Equivalent Incomes

One feature of the theory of fair allocation is that, like all branches of economic theory, it works with precise economic models describing specific distributive contexts. This is helpful and even necessary in order to grasp the specific issues of each allocation problem. The downside is that the theory offers many special solutions for particular contexts but is, for the time being, unable to give precise recommendations concerning the problem of devising a reasonable social criterion for a “real-life” context involving divisible and indivisible commodities, private and public goods, marketed and non-marketed goods and activities, production with unequal talents, multiple generations, and so on.

In the meantime, one can observe that many interesting solutions in the existing set of results are of the egalitarian-equivalent family. Such solutions apply the maximin criterion or the leximin criterion⁵¹ to specific numerical indexes of individual indifference curves or sets. Consider a given collection of nested sets $(B_\lambda)_{\lambda \in \mathbb{R}_+}$ such that $\lambda \leq \lambda'$ if and only if $B_\lambda \subseteq B_{\lambda'}$. If this collection contains sufficiently many sets, for every possible individual indifference set there is a set B_λ , called the “equivalent set,” that is tangent to the indifference set. In other words, the

⁴⁹ For a survey, see Hervé Moulin and William Thomson (1997).

⁵⁰ For a survey, see, e.g., Francois Maniquet (2007) or Fleurbaey and Maniquet (forthcoming).

⁵¹ The maximin criterion compares two vectors by their smallest component. The leximin criterion lexicographically compares the smallest component, the second smallest component, and so on.

maximum satisfaction the individual would obtain by choosing from B_λ corresponds to the given indifference set. The idea is then to evaluate individual situations in a given allocation by comparing their “equivalent sets” in the collection, i.e., the sets B_λ that would give them the same satisfaction (if they could freely choose from B_λ) as the current allocation. Such comparisons can be done unambiguously because these sets are nested: the larger, the better.⁵² The money-metric utility and the ray utility are two examples illustrating this “equivalence” approach: for the money-metric utility, consider the collection of sets B_λ defined by the condition $p^*x \leq \lambda$, where x denotes an individual bundle belonging to B_λ and p^* is the reference price vector; for the ray utility, consider the collection of sets B_λ defined by the condition $x \leq \lambda x^*$, where x^* is the reference bundle. Because such interpersonal comparisons can be made across individuals with different preferences, there is no serious obstacle to extending such social criteria to comparisons of social situations across different populations.

In order to illustrate how a particular criterion of the egalitarian-equivalent sort can be justified following the Arrovian approach, consider the social ordering axiomatized by Fleurbaey (2005) for a model with two goods, health H and income I (a composite commodity). Health is assumed to lie between zero and one. Individual utilities are assumed to be self-centered, i.e., each individual’s utility depends only on his own health and income. The social ordering that is axiomatized applies the maximin criterion to vectors of individual “healthy-equivalent incomes.” For any individual i , his healthy equivalent income is the level of income I^* that would make him indifferent between his current situation and a situation of perfect health with income I^*

(for monotonic preferences, this can also be described in terms of equivalent sets containing the bundles (H, I) satisfying the constraint $I \leq I^*$ and $0 \leq H \leq 1$). One can relate the axiomatics justifying this criterion to the Arrow axioms listed above in the following way: (1) Ordinalism is retained; (2) Independence is weakened, stipulating that the ranking of two allocations is unchanged when the utility levels and, in addition, the indifference sets at the two allocations remain the same with the new profile of utility functions; (3) Weak Pareto is retained; (4) A fairness axiom is introduced, stipulating that a Pigou–Dalton transfer reducing income inequality between two individuals having the same health yields an at least as good allocation, provided these two individuals have the same utility function overall, or have perfect health and the same utility function over perfect-health bundles.⁵³

A similar idea underlies the computation of equivalent income growth made in Gary S. Becker, Tomas J. Philipson, and Rodrigo R. Soares (2005). They consider a representative agent per country and compute equivalent income growth of a country as the growth of income that would have provided the same gain in satisfaction if health (life expectancy, in their work) had remained constant. This amounts to taking initial health, instead of perfect health, as the reference for computing the equivalent income. This is a quite natural and intuitive measure, which can be viewed as a generalization of the idea of equivalent variation.⁵⁴ The standard notion of equivalent variation takes initial prices as the reference for the

⁵² I assume the absence of satiation.

⁵³ The restriction to individuals with same utility function makes the axiom weaker and is motivated by the fact that an individual with a greater income at the same health level than another is not necessarily better off if he has different preferences (e.g., he may suffer more from this health condition).

⁵⁴ This kind of generalization was studied, in particular, by Hammond (1994).

computation of equivalent income change. The price vector is just one among many variables, such as personal health or features of the environment, which influence indirect utility. It makes sense to compute the income variation that would have generated the same variation in satisfaction if all variables other than income had been kept constant. It is not easy, however, to incorporate equivalent variations into a consistent social criterion, especially if one wants to avoid the representative agent approach. In particular, equivalent variations are not convenient for interpersonal comparisons as they focus on changes rather than levels. One cannot, for instance, compare the final equivalent incomes, computed with the Becker et al. approach, for two individuals having different initial health. Another problem with a variable reference is that the chained index may have little meaning: the addition of equivalent income gains with moving references is not equal to the augmented income gain with respect to a fixed reference. In spite of these difficulties, one may consider that a systematic computation of such equivalent income growth (if possible, by adding up individual equivalent variations rather than assuming a representative agent) would provide valuable information in complement to ordinary income growth. Comparing the two curves of growth rates would be informative about the evolution of the dimensions of quality of life incorporated into the computation. This can be justified by a variant of the decomposition formula (5). Consider a social welfare function $W(u_1(m_1, z_1), \dots, u_n(m_n, z_n))$, where m_i denotes income and z_i the other dimensions. Let $\nabla_z u_i$ denote the gradient of u_i with respect to z_i and let willingness-to-pay (equivalent variation) be defined as:

$$WTP_i = dm_i + \frac{1}{\frac{\partial u_i}{\partial m_i}} \nabla_z u_i dz_i.$$

One can then write, in similar fashion as in (5):

$$dW = \sum_i \beta_i WTP_i = \bar{\beta} \sum_i WTP_i + \sum (\beta_i - \bar{\beta}) \left(WTP_i - \frac{1}{n} \sum_j WTP_j \right),$$

where $\beta_i = (\partial W / \partial u_i)(\partial u_i / \partial m_i)$, $\bar{\beta} = 1/n \sum_{i=1}^n \beta_i$.

Obviously, the danger in computing the first term of the decomposition without the second is to distort perceptions about social welfare. This issue is reminiscent of the difference between old and modern cost-benefit analysis (CBA) and it is worth mentioning here how CBA could benefit from calibrating individual utilities in terms of equivalent incomes. The traditional type of CBA, based on compensation tests, compensating and equivalent variations, and sums of WTP_i , has repeatedly been criticized by welfare economists for being biased in favor of the rich (whose willingness-to-pay is greater in the presence of income effects) and for being susceptible to inconsistent conclusions.⁵⁵ A more sophisticated form of CBA, relying on a social welfare function, consists of substituting a weighted sum $\sum_i \beta_i WTP_i$ to the old-fashioned unweighted sum, with weights $\beta_i = (\partial W / \partial u_i)(\partial u_i / \partial m_i)$ measuring the social marginal value of income of individual i .⁵⁶ Practitioners have generally been embarrassed about the choice of weights in applications, for lack of a natural measure of marginal utility, and have often taken some decreasing function of income. The problem appears more tractable if one takes a social welfare function $W(u_1(m_1, z_1), \dots, u_n(m_n, z_n))$ in which $u_i(m_i, z_i)$ is i 's equivalent income,

⁵⁵ See, e.g., Arrow (1951), Robin W. Boadway and Neil Bruce (1984), Blackorby and Donaldson (1990).

⁵⁶ See Jean Drèze and Nicholas Stern (1987), Richard Layard and Stephen Glaister (1994).

i.e. the level m^* such that i is indifferent between (m_i, z_i) and (m^*, z^*) , where z^* is a reference value for nonincome variables. Then the only ethical choices that remain to be done concern the reference z^* and the function W (in particular its degree of inequality aversion).

Like equivalent variations, the notion of equivalent income can be applied to any number of dimensions of quality of life. One simply has to fix a reference vector for these dimensions, and compute the equivalent income as the income that would yield the same satisfaction if quality of life were at the reference level. Equivalent income equals ordinary income minus the willingness to pay to have quality of life at the reference level. Considering market prices as an element of quality of life, one obtains the money-metric utility and a rigorous way to perform PPP corrections of income. Considering household size as an element of quality of life for individuals, one can compute the equivalent income that would give the member of a household the same individual satisfaction if he belonged to a reference type of household (e.g., if he were single). This method, proposed for instance in Martin Browning, Pierre-Andre Chiappori, and Arthur Lewbel (2006), appears superior to standard household equivalence scales that are not identifiable from market demand in absence of heroic assumptions.⁵⁷ Considering the risk of unemployment, it is natural to compute the certainty equivalent income by deducting a risk premium from ordinary income. Considering leisure, there are several reason-

able possibilities highlighted in the theory of fair allocation, not all of which involve computing an equivalent income. A simple option is to take a reference quantity of labor.⁵⁸ In order to compare the standard of living across OECD countries, Fleurbaey and Gaulier (forthcoming) have used the equivalent income approach (with estimations based on macroeconomic data) for the corrections concerning leisure, health, the risk of unemployment, and household size.

The equivalent income approach involves a special case of the collection of nested equivalent sets that are used in the "equivalence" approach. Other options exist and appear no less reasonable. Instead of fixing the non-income variables at a reference level, one could make them vary from one set B_λ to the other along a progression path in which quality of life and income would grow simultaneously.

4.4 Criticisms

Money-metric utilities, as mentioned in subsection 2.5, have triggered criticism that can also be directed at the equivalence approach in general.

The first criticism points to the neglect of welfare aspects, in particular those having to do with special needs (related for instance to health or disabilities). As stressed in Sen (1985), resourcist approaches may seem fetishistic about external resources, neglecting the fact that different individuals have unequal abilities to transform resources into flourishing. Needs are, however, easily

⁵⁷ For a synthesis on household equivalence scales, see Lewbel (1997). Equivalence scales based on individual equivalent income may ultimately provide familiar formulae. It is shown in Fleurbaey and Guillaume Gaulier (forthcoming) that, if one assumes that households spend half of their budget on local public goods and the rest on private goods and that individuals are identical within households, equivalence scales based on individual equivalent income coincide with the OECD household equivalence scales relying on the square root of household size.

⁵⁸ In the egalitarian-equivalent family, there are three other salient options, corresponding to different ethical principles. One option consists in computing a tax-free equivalent budget (i.e., the hypothetical wage rate which would maintain satisfaction in absence of transfers). Another option consists in computing an equivalent budget at the market wage rate for unskilled work. A third option involves computing an equivalent budget corresponding to a zero net wage rate (with monotonic preferences, this boils down to computing a leisurely equivalent income). For a survey, see Fleurbaey (2008).

incorporated into the equivalent income approach if they are explicitly taken into account among the dimensions of quality of life. Therefore, properly handling needs in interpersonal comparisons does not require abandoning Ordinalism but merely broadening the object of ordinal preferences in order to encompass certain personal parameters. This procedure supposes that individuals have well-defined preferences over various possible levels of needs and over various combinations of needs and income. Although this assumption may appear demanding, it seems hard to avoid relying on this sort of preferences if one seeks to determine how to compensate specific needs with transfers of external resources. It should also be emphasized that the egalitarian-equivalent approach is not really tied to a narrowly materialistic notion of resources, as it can be applied to any broader class of objects of preferences, such as, for instance, functionings or capabilities. What characterizes the approach is its method of comparing indifference sets, not the space in which the indifference sets are defined. Beyond the incorporation of special needs or broader notions of functionings into the objects of preferences, would it be advisable to take account of more subjective dimensions of well-being, such as happiness or levels of satisfaction? This is obviously a controversial issue, which will be examined in greater detail in section 6.

The second criticism is that a social welfare function W whose arguments are individual money-metric utilities may fail to be quasi-concave in resources (Blackorby and Donaldson 1988). This observation can, in fact, be substantially generalized on the basis of results from the theory of fair allocation (e.g., Maniquet and Yves Sprumont 2004): Any approach that obeys the Pareto principle and that evaluates individual situations by the indifference sets at the contemplated allocation (i.e., not just money-metric utilities) will fail to be quasi-concave unless it gives absolute priority to the worse off. The literature on fair allocation deduces from this observation that the maximin and leximin criteria are the only acceptable aggregators. This may sound an ominous result because these criteria are considered extreme on the spectrum of inequality aversion, but the theory is flexible about how to evaluate individual situations. For instance, it can justify the *laissez-faire* policy if individual endowments are considered fair and are suitably incorporated into the comparisons of individual indifference sets.⁵⁹

A third criticism has been raised against money-metric utility functions. Developed by Slesnick (1991) and inspired from Kevin Roberts (1980), the point is that money-metric utility functions depend on a reference price p^* and that, if one wants the social welfare function to be independent of these

⁵⁹ See Fleurbaey and Maniquet (forthcoming). An alternative reading of this result is that if one does not want to adopt the maximin or leximin criterion, one has to accept certain apparent violations of inequality aversion. Consider the healthy-equivalent income, and imagine two individuals with health equal to 0.5 (recall that the range for health is $[0,1]$). Suppose that their ordinary incomes in allocation x are (100,200) and their equivalent incomes (80,150). Imagine that a regressive transfer between them would produce a new situation with ordinary incomes (90,210) and equivalent incomes (78,190). If one considers that, for two healthy individuals, the distribution (78,190) is better than (80,150) because the large gain to the better-off outweighs the small loss to the worse-off, it is only

logical, by Pareto-indifference, that for the half-healthy individuals under consideration, the regressive transfer yields an improvement. In this way, a positive but finite inequality aversion over equivalent incomes may imply a lack of inequality aversion about the distribution of income among unhealthy individuals. This is intuitively defensible in this example if one observes that the regressive transfer triggers a reduction in both individuals' willingness to pay for a good health, especially for the beneficiary of the transfer. This example may also suggest that, in practice, such violations of the Pigou–Dalton principle are unlikely to occur in dramatic proportions if the aggregator has substantial inequality aversion and health is a normal good with moderate income effects.

reference parameters, severe restrictions are required (e.g., homothetic preferences). Similarly, the computation of equivalent incomes is sensitive to the choice of the reference. It is not clear, however, why one should worry about such dependence, or impose independence from the reference, because the choice of a reference need not be arbitrary. The example of the healthy-equivalent income, for instance, suggests that the reference in this case is natural. With the healthy-equivalent income, one can compare individual situations in perfect health solely in terms of income, without inquiring about individual preferences about health (because equivalent income is then equal to ordinary income), whereas situations with less than perfect health require taking account of individual preferences about health and income. This seems the proper methodology, and taking a low level of health as the reference would have counterintuitive implications, with the sick being compared without consulting their preferences while the healthy could not be simply compared in terms of income. Similarly, as mentioned above, the literature on fairness suggests specific references for the case of leisure. The literature on equivalence scales has generally taken singles as the reference type of household, and again this seems rather natural, although perhaps debatable. Even in the case of reference prices for PPP computations, it would be an exaggeration to claim that all price vectors are equally acceptable candidates for the reference. In sum, even if the theory of references is still in its infancy, the idea that the choice of the reference is arbitrary is simply unwarranted.

Another issue has to be mentioned. The difficulty of estimating individual preferences, an essential piece of data for any application of the approach, should not be underestimated. There are three possible sources of data on preferences. First, preferences can be revealed by observed choices. This is the standard source of data on

preferences for market commodities, but it is not reliable for nonstandard goods (such as working conditions) and is obviously important for preferences about nonchosen aspects of quality of life. Second, there are stated preferences which can be obtained from contingent valuation questionnaires (in which respondents are directly asked how much they would be willing to pay for such or such change in their situation or in the environment) or from discrete choice experiments (in which respondents are randomly offered menus and asked to choose their preferred option). Contingent valuation studies relative to remote features of the environment have been highly debated.⁶⁰ Although one may hope that for their personal situations people do have more consistent preferences, the accumulation of empirical evidence about bounded rationality and poor decision-making ability appears worrisome.⁶¹ A third possible source of data consists in satisfaction surveys, which have been used by several authors to estimate willingness-to-pay for various aspects of quality of life, but are likely to be rather noisy because unobserved individual characteristics influence responses to such questions.⁶² Therefore, the theoretical possibility to aggregate individual preferences into a consistent social criterion needs to be supplemented with an effort in the methods of estimation of preferences. It might also be worth thinking more carefully about whether one should be interested

⁶⁰ Daniel Kahneman, Ilana Ritov, and David A. Schkade (1999) have argued that, in such contexts, people express symbolic attitudes, not economic preferences. This shows, in particular, in the fact that willingness-to-pay is insensitive to the quantities at stake, for instance the number of birds saved from pollution. See also Peter A. Diamond and Jerry A. Hausman (1994).

⁶¹ B. Douglas Bernheim (2008) reviews the relevant literature and suggests that at least some part of individual preferences is stable and consistent, so that welfare economics can work with this part (formally, individual preferences should then be formalized as acyclic partial orders rather than transitive complete orders).

⁶² See section 6.

in people's immediate preferences (likely to be volatile and inconsistent) or in some form of deeper preferences, which may be much harder to elicit.

In conclusion, money-metric utilities and equivalent incomes, linked to the equivalence approach from the theory of fairness and to social choice theory, might be a promising field for research and applications. The approach has so far been applied mostly to special economic models dealing with particular allocation problems. A rigorous analysis of the definition of the sets B_λ in a richer context remains to be done, but intuitive applications in terms of equivalent income are quite easy to conceive.

5. Social Indicators

I now turn to nonmonetary approaches. There is now a wealth of summary indicators that combine various domain indicators of economic, social, and environmental performance.⁶³ Prominent indicators include the Human Development Index (HDI) and the indicators computed by Lars Osberg and Andrew Sharpe (2002) and Marc Miringoff and Marque-Luisa Miringoff (1999).

Discussing the theory underlying such indicators can be done shortly because there is little theory. The weights of the various domain indicators in the general index are conventional and the proponents of such indicators rarely formulate a framework for a rational discussion about what these weights should be.⁶⁴ One can of course invoke the ethical preferences of the observer and ask her, for instance, how she trades the suicide rate off against the literacy rate, but there is little philosophical or economic theory

that gives us clues about how to form such preferences.

A problematic feature of these indicators is that they are not individualistic. That is, they do not form an aggregate of individual indexes but instead add up social indicators for various domains of individual well-being. Consider for instance health and income. Adding up a social index of health (e.g., average life expectancy at birth) with a social index of income (e.g., GDP), as is done in the HDI together with an index of education, gives the same result for societies having the same value for both indexes, independently of the correlation between health and income at the individual level. If one thinks that such a correlation is important, because a society with no correlation is better than a society in which being richer implies being healthier, one cannot rely on these summary indicators.⁶⁵ Assuming that information about this correlation is available, one can of course use it as an additional component of the summary indicator, but it appears a much more satisfactory procedure to evaluate situations at the individual level before going up to social summaries.⁶⁶ Obviously, it must be acknowledged here that, in absence of individual data making it possible to estimate the joint distribution of the various aspects of individual situations, such indicators can play a useful role. It is already an achievement to have comparable statistics on income and longevity for almost all countries and an index like the HDI may be worth keeping alive before better data become available.

Steve Dowrick, Yvonne Dunlop, and John Quiggin (2003) propose an economic theory for the use of such indicators, based

⁶³ For a survey, see Jean Gadrey and Florence Jany-Catrice (2006).

⁶⁴ This point is made, e.g., by Martin Ravallion (1997) commenting on the HDI.

⁶⁵ See Indranil Dutta, Prasanta K. Pattanaik, and Yongsheng Xu (2003).

⁶⁶ An additional issue often mentioned about the HDI is that, for developed countries at least, the other components than GDP move very little so that the movements of the index mostly reflect growth rates.

on a revealed preference argument. Their approach supposes a representative agent (which of course puts aside the correlation issue) facing a technology for the production of performance in the various domains, together with a budget constraint. They propose to say that country *A* is better off than country *B* if, given its technological and budgetary constraints, country *A* could obtain a vector of performances dominating the vector of *B*. This approach enables them to make binary comparisons. It is in principle possible to find that *A* is better off than *B* and *B* is better off than *A*, simultaneously (as occurs in their data for Finland and Austria). They also discuss the possibility of computing numerical indexes by assuming homothetic preferences over commodities for the representative agent, with identical preferences across countries. While this approach may yield interesting insights about comparisons of multidimensional feasible sets in the space of domain indicators, it appears difficult to connect it to a standard notion of social welfare. The difficulties in estimating the technology of production of social performance in various dimensions (Dowrick and coauthors only look at income and health) may also render the approach hardly applicable on a large scale.

6. Happiness

The literature on happiness has surged in the last decade. The findings are well summarized in many surveys.⁶⁷ The question that has to be examined here is how this approach can inform the evaluation of social welfare. Are we close to substituting GNH ("Gross National Happiness") for GDP? No author argues that happiness studies are

of no value at all, but there is wide variety of positions, from those who propose to measure and maximize national happiness (Diener 2000, Kahneman et al. 2004, Layard 2005) to those who firmly oppose this idea on various grounds (Tania Burchardt 2006, Frey and Stutzer 2007, Martha C. Nussbaum 2008). There seems to be a consensus on the idea that happiness studies suggest a welcome shift of focus, in social evaluation, from purely materialistic performances to a broader set of values. In particular, the importance of social relations is highlighted by this literature while the role of consumption is downplayed.

6.1 Subjective Well-Being is Measurable

Above all, one can consider that the traditional suspicion among economists about the possibility to measure subjective well-being (SWB) should be assuaged by the recent progress. Combining questionnaire surveys ("Taking all things together, how satisfied are you with your life as a whole these days? Are you very satisfied, satisfied, not very satisfied, not at all satisfied?"), the methods of experience sampling (mood declaration at random moments of the day) and day reconstruction (short recall mood declaration), physiological measures (hormone concentration, skin conductivity, . . .), neurological measures (brain activity), as well as behavioral observations (smiling . . .), it now seems more possible than ever to obtain a rather consistent estimate of SWB. Doubts remain about whether the results coming from verbal statements and scaling questions can really be compared across individuals (especially across cultures),⁶⁸ but it would be an exaggeration to give no

⁶⁷ See, in particular, Ed Diener (1994, 2000), Diener et al. (1999), Bruno S. Frey and Alois Stutzer (2002), Kahneman and Alan B. Krueger (2006), Layard (2005), Andrew J. Oswald (1997).

⁶⁸ For instance, Krueger et al. (2008) find out that American and French women seem to have different attitudes about presenting themselves as "very happy" or "happy."

significance to interpersonal patterns in the data. On the whole, one can be optimistic about the increasing reliability⁶⁹ of data on SWB in the near future.

One important fact that psychological studies reveal, and which is sometimes downplayed in the economic part of the literature, is the multidimensionality of SWB (Diener et al. 1999). A key divide opposes cognitive evaluations (what people think of their life) to affective, emotional states (how they feel in their life). The affects themselves come in many shapes and colors, with a surprising independence between the positive and the negative affects which seem to be connected to different mechanisms in the brain.⁷⁰ In this light, Bentham's and Edgeworth's "utility" is an artificial concept that ignores the complexity of human psyche. One shortcoming of the available set of findings is the relative lack of separation between affects and judgments in many studies. Questions about satisfaction with life "as a whole" are disturbingly influenced by the mood of the day, the current weather, or incidental events that temporarily shift someone's mood such as finding a dime just before answering the questionnaire. Recent "ladder-of-life" questionnaires asking respondents to rank their life on a scale between 0 (worst possible life) and 10 (best possible life) may come closer to identifying the cognitive part of people's satisfaction (Deaton 2008). Even if instantaneous mood is the result of sensations combined with judgmental evaluation, one may be more confident that specific surveys on emotional states do capture affects.

Kahneman and Krueger (2006) propose to synthesize data on moment-to-moment emotions into a "U-index" that measures the proportion of time in which the predominant affect is unpleasant.⁷¹ This index can be computed for individuals or averaged over groups; it can also be computed for particular activities or for the whole day. An advantage of this index is that it is cardinal (it makes sense to say that an individual has 20 percent more unpleasant time than another) without requiring a cardinal calibration of emotional scales (except the rather robust distinction between positive and negative affects). In counterpart, it does not capture variations in the intensity of unpleasantness and ignores what happens in the large proportion of the time in which the dominant affect is positive.

Much of the literature in happiness studies seeks to understand the determinants of happiness, with many fascinating findings about the relative importance of income, health, social status and unemployment, marital status and family life, religions, rights and political freedom, and about the complexity of dynamic effects and comparisons to peer groups. A particular difficulty that pervades many studies is that cross-section analysis tracks correlates of happiness but the direction of causation is not always clear. For instance, a strong correlation between unemployment and unhappiness may reflect the effect of unemployment on happiness but also the underlying combined influence of a third factor such as personality on job stability and happiness or even a direct influence

⁶⁹ A balanced evaluation of the reliability of subjective well-being data is made by Krueger and Schkade (2008).

⁷⁰ Classifications of affects along two or more dimensions (e.g., positive/negative valence and high/low activation, as proposed by James A. Russell 1980) or in terms of discrete categories (Paul Ekman 1992) have triggered many debates in psychology. No simple classification seems able to distinguish all emotions (for instance, anger and fear are both intense negative emotions). On these issues, see Nico H. Frijda (1999).

⁷¹ As explained in Kahneman and Krueger (2006), doubts about comparability of answers across individuals motivate the difference between the "U-index" and the index initially proposed in Kahneman et al. (2004), which was a time-weighted mean of the average happiness obtained by the population in various activities. The latter required a cardinal scale of affect whereas the former only needs to identify moments where a negative affect dominates.

of mood on the willingness to quit one's job.⁷² The causality issue, however, does not undermine the accuracy of the measures of SWB and has no direct implication on their potential normative relevance, to which I now turn.

6.2 Treadmills

The fact that SWB *can* be measured does not imply that it *ought* to be taken as the metric of social evaluation. It is rather surprising that the literature on happiness refers very little to the lively philosophical debates of the previous decades about welfarism and, in particular, the criticisms raised by Rawls (1982) and Sen (1985) against utilitarianism.⁷³ Nonetheless, one of the key elements of that earlier debate, namely, the fact that subjective adaptation is likely to hide objective inequalities, does happen to come up as a puzzle challenging happiness specialists.⁷⁴ Subjective well-being seems relatively immune in the long run to many aspects of objective circumstances, individuals displaying a remarkable ability to adapt. After most important life events, satisfaction returns to its usual level and the various affects return to their usual frequency.⁷⁵ A key determinant of usual mood and satisfaction is the individual's personality (extroversion versus neuroticism).

⁷² Many authors, such as Frey and Stutzer (2002) or Ada Ferrer-i-Carbonell and Frijters (2004), recommend using panel data so as to eliminate individual fixed effects. Andrew E. Clark, Frijters, and Michael A. Shields (2008) discuss the limits of panel data. Controlled or natural experiments could be helpful as well.

⁷³ Two exceptions are Burchardt (2006) and Erik Schokkaert (2007a). Layard (2005) also mentions and quickly rebuts some of the arguments against welfarism.

⁷⁴ The first influential study on adaptation (Philip Brickman and Donald T. Campbell 1971) even predates Sen's example of the tamed housewife who accepts her impoverished condition.

⁷⁵ An individual's aspirations do not only depend on past experience but also on the achievements of his group of reference (see, e.g., Clark and Oswald 2002, Clark, Frijters, and Shields 2008, Bernard Van Praag and Ferrer-i-Carbonell 2007, Shane Frederick and George Loewenstein 1999).

If SWB is not so sensitive to objective circumstances, should we stop caring about inequalities, safety, and productivity?

Kahneman (1999) wonders whether this phenomenon should be attributed to a hedonic treadmill reflecting a progressive desensitization of subjects to repeated stimuli or to an aspiration treadmill due to the fact that the subjects evaluate their situation according to adaptive aspiration levels. This is important for normative conclusions. Suppose, following the early utilitarian tradition, that the normatively relevant metric for social evaluation is experiential hedonic utility. If it turned out that adaptation impacts aspirations more than hedonic experience, then one could safely seek to measure a hedonic index of "national happiness" because such an index would actually be sensitive to objective circumstances. Such an index could not be based on declarations of satisfaction and more direct observations of emotions would be necessary.⁷⁶

Kahneman and Krueger (2006), however, write that preliminary evidence actually points to the predominance of the hedonic treadmill over the aspiration treadmill because data on satisfaction show a greater correlation with life events than records on affects. This undermines the idea of measuring national happiness in terms of hedonic utility, at least if one would like such a measure to display a certain sensitiveness to material life conditions—the alternative is to declare material conditions to be of little importance. Nonetheless, even if less overwhelming, the aspiration treadmill also questions using reported satisfaction as a measure of well-being. In a recent study of

⁷⁶ Another obstacle to the measure of hedonic experience that is described by Kahneman, Peter P. Wakker, and Rakesh Sarin (1997) is that people's memory keeps track of experienced affects in a distorted way, giving excessive attention to peaks and last moments of a given episode. One then has to find ways (such as the experience sampling and the day reconstruction methods) to retrieve individuals' true hedonic experience.

ladder-of-life surveys over the world, Deaton (2008) observes that globalized information about living conditions makes it unsurprising that respondents rank their life in accordance with common rankings of living standards and argues that such surveys cannot be trusted over time because of shifting references. Moreover, he concludes that “neither life satisfaction nor health satisfaction can be taken as reliable indicators of population well-being, if only because neither adequately reflects objective conditions of health” (p. 70).⁷⁷

6.3 *Two Conceptions of SWB*

The early debate on liberalism versus welfarism and the recent literature on happiness suggest that two normative conceptions of how to make use of subjective data compete.

The welfarist conception, inspired by Bentham and Edgeworth, predominates in the happiness literature even when it is not explicitly or fully endorsed. In its hedonic version, it views utility primarily as an affect characterized by two quantities, duration and level, the latter being measured on a numerical scale with negative values for unpleasant affects and positive values for pleasant ones. As positive and negative affects can coexist during the same short episode, the instantaneous utility level is conceived as a net balance sheet. Assuming separability over time, total utility can be computed as the integral of instantaneous utility over time. As individuals do not recall this integral efficiently and are often mistaken about what generates positive and negative affects, their preferences and decisions fail to maximize their “true” utility. For instance, they think that a new car will make them happy but, after a couple of weeks,

their utility increment dwindles. They think that working extra hours and earning extra money will make them happy but their impoverished family relationships turn out to have a much more lasting effect on their utility. If one wants to maximize national happiness, one will then have to conceive policies that will possibly curb the satisfaction of people’s preferences in order to promote their true happiness. Hedonic adaptation, however, suggests that psychological interventions may be more efficient than most changes in objective life conditions. In this perspective, researching a happiness drug without negative side effects might even be a more urgent priority than anything else.⁷⁸

A variant of this conception puts more stress on the cognitive part of happiness. Satisfaction judgments may be argued to matter more than hedonic utility because affects are just one domain of life, and most people are willing to trade hedonic affects for objective achievements.⁷⁹ People also differ in the kind of affects they want to cultivate. In particular, the hedonic model of utility level on a linear scale does not square well with the observation that many people prefer quiet contentment to intense joy.⁸⁰ This variant seeks to find a measure

⁷⁸ This position is endorsed by Layard (2005), who defends it by arguing that if such a drug existed, most people would take it. But observe that for deciding to take the pill (assuming taking the pill does not hinder other activities), people only have to give some value to their own happiness, not to give it exclusive value.

⁷⁹ The popular saying, often attributed to Aristotle, that happiness is the only ultimate, noninstrumental goal of human beings is either tautological (when happiness is identified to the satisfaction of goals) or empirically incorrect (when happiness is an affect). On the views of Aristotle, Mill, and Bentham on this issue, see Nussbaum (2008).

⁸⁰ On this point, see Diener (2000, p. 36), who also insists in this paper on the multidimensionality of subjective well-being (SWB) and proposes an ecumenical set of indicators: “Ideally, the national SWB indicators would include various components of SWB, such as pleasant affect, unpleasant affect, life satisfaction, fulfillment, and more specific states such as stress, affection, trust, and joy” (p. 40).

⁷⁷ A puzzling finding in Deaton (2008) is indeed that life satisfaction correlates well with (the logarithm of) income in cross section but not with the level of health, not even HIV prevalence. “It seems astonishing that reported life satisfaction should be unaffected by a plague whose severity is unparalleled in modern times” (p. 63).

of satisfaction that can be used as a measure of utility and be compared and aggregated across individuals. It is also interested in finding the determinants of satisfaction and to fight people's misconceptions of what really gives them a greater satisfaction in the long run.

An alternative to welfarism, in its hedonic and satisfaction versions, can be imagined on the basis of the approach of liberal philosophers such as Rawls, Dworkin, and Sen. Like satisfaction welfarism, it recommends paying attention to individuals' cognitive evaluation of their life and would, for instance, seek to enhance affects only in proportion to their importance in individuals' view of a good life.⁸¹ But, unlike satisfaction welfarism, this approach rejects the idea that one should compare satisfaction across individuals and seek to maximize some aggregate of individual satisfaction levels. The key objection from the liberal approach against satisfaction welfarism, perhaps, is that satisfaction welfarism fails to distinguish "obtaining what one wants" from "being satisfied" and mistakenly focuses on the latter instead of the former. There are three ways to be satisfied, and obtaining what one wants is only one of them. One can also become satisfied by adapting one's aspiration level or by adapting

one's preferences.⁸² If we could move to a situation in which people would have less of what they want but would reduce their aspiration levels and end up being more satisfied, the welfarist approach would approve this change and the liberal approach would oppose it.⁸³

A good illustration of this point and of the different implications of the two conceptions is provided by how they interpret the "Easterlin paradox."⁸⁴ For the welfarist approach, the paradox reflects the fact that individual decisions are misguided. People strive for material achievements that have little impact on their SWB, whether it is measured in terms of hedonic utility or in terms of satisfaction. For the liberal approach, there is no paradox at all: people may rationally want to improve their material conditions of life even if, in the long run, it does not increase their hedonic experiences (because of the hedonic treadmill) and does not much improve their evaluation of life success (due to the aspiration treadmill). Both *ex ante* and *ex post* they do strongly prefer the higher living standard and would not want to stay at or go back to the lower level. They may perhaps

⁸¹ The liberal approach can accept the idea that people sometimes make bad choices. But the issue, from this perspective, is not mainly that people fail to accurately predict experienced affects—in fact, if people have systematically distorted memories and predictions of hedonic utility, one could argue that it undermines the value of promoting hedonic utility (Daniel M. Hausman 2007). Rather, mistakes reflect the discrepancy between immediate preferences and deeper preferences due to a lack of information and deliberation in daily circumstances. For this reason, the liberal model is actually compatible with a kind of soft paternalism in which some correction to individual preferences is permitted in the computation of social welfare provided one is confident that it produces a better picture of people's deep preferences.

⁸² Brian Barry (2008) compares an individual who would seek to be satisfied *per se*—instead of getting what he wants—to a football fan who would support whatever team is most likely to win. What kind of football fan would that be?

⁸³ The debate could be pursued further. The welfarist can bite the bullet and argue that adapting to one's current condition is an honorable way to become satisfied, which should be encouraged. The liberal would perhaps retort that there is a difference between elevating one's goals beyond basic materialism and adopting bland ambitions.

⁸⁴ This paradox, as proposed first by Richard A. Easterlin (1974, 1995), was that, in developed countries, average happiness appears relatively insensitive to income both in cross section and in the long run. More complete recent data have invalidated the cross-section version of the paradox, as average satisfaction with life and the logarithm of income per capita seem aligned even for rich countries (Deaton 2008; Betsey Stevenson and Justin Wolfers 2008). The diachronic version of the paradox is debated (in addition to the two previous references, see, in particular, Krueger 2008 and Clark, Frijters, and Shields 2008).

overestimate the subjective benefits of material growth, but this is unlikely to distort their decisions so much that they would not seek material improvements otherwise. Assuming that they do value material achievements such as the ability to travel and communicate or a better control of health, they would still strongly prefer the higher standard of living even if they perfectly predicted the stability of their affects and of their satisfaction. In other words, such stability is compatible with people having definite preferences about the content of their lives. This is Frederick and Loewenstein's (1999) conclusion of a long study of adaptation: "Assuming that future research provides a deeper understanding of hedonic adaptation, is it likely that such information would cause people to conduct their lives differently? Would they stop wearing seatbelts with the assurance that they would get used to being paralyzed? Would they exploit an embezzlement opportunity knowing that prison wouldn't be all that bad in the long run? We suspect not (p. 320)."

The two conceptions also diverge a little on the analysis of the inefficiency of growth in connection with peer comparison and positional goods. For the welfarist approach, it is clearly inefficient to want to earn more if this triggers a rat race that ultimately decreases utility. For the liberal approach, one must distinguish two phenomena. If individual preferences are concerned with positional goods, society will indeed be trapped in a prisoner's dilemma in absence of coordination. People will sacrifice their personal comfort in order to improve their relative position, and this is collectively wasteful.⁸⁵ No additional inefficiency, however, is entailed in this approach by the fact that aspiration levels are influenced by the others' achievements. Because efficiency must be assessed with respect to

the individual ordinal preferences, not with respect to satisfaction levels which fluctuate with aspirations.

6.4 *A Revolution in Utility Theory?*²

The developments in happiness studies may sound as if the standard model of well-being based on a well-defined utility function (or preference ordering) bearing on personal consumption, which inspires in particular the monetary measures reviewed in sections 2–4, has to be radically revised. The multidimensionality of SWB, the hedonic and satisfaction treadmills, and the importance of social comparisons appear to require a much more complex picture. Paradoxically, this questioning is not prominent in happiness studies, where many authors seem to keep searching for the "utility" Graal and to believe that the recent developments bring them closer to it.

As a matter of fact, the required changes to the standard model may be less dramatic than it appears. Suppose one reads the classical model of consumer preference as describing what the individual wants for his life, and let us take the object of preference to be a comprehensive vector of "functionings" or "capabilities" (see section 7) rather than a narrow vector of consumption. Such preferences correspond presumably more to the cognitive part of SWB than to the affective part. Hedonic utility, in this model, is just one component of the object of preference,⁸⁶ and the hedonic treadmill means that this component is relatively stable and immune to changes in the other components of the life vector. It does not imply any problematic instability in the preference ordering, even if it may sometimes produce apparent changes

⁸⁵ The disincentive effects of income taxation, for instance, can be viewed favorably as a brake on wasteful competition for positional goods (Layard 2005).

⁸⁶ Becker and Luis Rayo (2008) and Loewenstein and Peter A. Ubel (2008) propose to consider subjective well-being as an argument of utility, alongside more objective arguments.

in preferences. For instance, if one initially dislikes a particular activity and eventually finds great fun in it, one may come to like it because it has a different impact on hedonic utility. This is compatible with a stable underlying ordering over hedonic utility and other dimensions.

Whether the same can be said about the satisfaction treadmill is more ambiguous. An adaptation of the aspiration level and of the references for comparisons is compatible with a stability of the preference ordering. It is possible to become, say, fussy about comfort and safety because one gets used to affluence, without changing one's ranking of the various possible life vectors. But the satisfaction treadmill may also partly include an adjustment in the orientation of ordinal preferences. Someone who initially prefers the countryside may come to like urban life after several years in a town. Most of such changes in preferences can be interpreted in terms of learning rather than a deep alteration of the direction of tastes. But even real changes in preferences are not seriously challenging for the classical model of preferences because it would be an exaggeration to interpret this model as requiring perfect stability of the ordering. Recall that there are measures of well-being, such as money-metric utilities, which permit comparisons across different preferences.

Perhaps the most challenging part of happiness studies, for the classical model, is the revealed importance of social comparisons.⁸⁷

⁸⁷ The importance of social comparisons and interdependent preferences has also been the hallmark of a large recent literature in experimental behavioral economics, surveyed in Ernst Fehr and Klaus M. Schmidt (2003) and Joel Sobel (2005). This line of research has shown the importance of reciprocity and concerns for fairness in explaining strategic interactions. Although welfare economics should definitely draw on these findings in order to have a realistic picture of individual preferences, the problem we examine here is different: how should one evaluate and compare individual situations when individuals care about others or about their relative standing?

Much of traditional welfare economics is built on the assumption that what matters is what individuals want for themselves. What they want for their neighbors has been the subject of some perplexity. Many models in welfare economics have simply assumed that individuals are selfish (i.e., only have self-centered preferences), which eliminates the issue in a way that now appears very unrealistic. Another possibility, which may have implicitly underlied the theory of fair allocation, is to declare that the non-self-centered part of individual preferences is irrelevant and should be disregarded. This makes a lot of sense because it nicely separates the question of personal interests, for which individual preferences are the best guide according to consumer sovereignty, from the question of social conflict, for which some different kind of ethical preference has been constructed on a more solid basis than the contingent benevolent and malevolent feelings of the population. One problem with this strategy of separation is that the self-centered part of individual preferences may not be well-defined in isolation and may substantially depend on the others' consumption. Whether such social influences make the self-centered preferences too unreliable for classical welfare economics may be considered an open issue, although no author seems to be taking that line. After all, it is still possible to take self-centered preferences as the measure of individual well-being even if they strongly depend on the whole allocation. But this, once again, requires a measure of individual well-being that cuts across different preferences. Otherwise it becomes problematic to say whether an individual is better off in an allocation than in another, because his self-centered preferences are different in the two allocations. Even more challenging is the fact that delineating the self-centered part of preferences may itself be difficult. Preferences over positional goods and over

one's relative position seem to irreducibly combine a self-centered part and a social part.⁸⁸

In conclusion, recent developments in happiness studies do not seem to require an overhaul in the classical utility model of welfare economics but call for a more serious apprehension of the influence of social comparisons on, and their place in, individual self-centered preferences.

6.5 *Interpersonal Comparisons and Inequalities*

The authors who propose to maximize average happiness seem unconcerned with inequalities between individuals. But it is conceptually easy to introduce inequality aversion in the definition of national happiness by putting a greater weight on the less happy and the more miserable individuals, as suggested by Layard (2005).⁸⁹

It is interesting to contrast the two approaches distinguished in subsection 6.3 on how they deal with interpersonal comparisons for the purpose of social evaluation. The welfarist approach advocates comparing people either in terms of experiential utility (hedonic welfarism) or in terms of satisfaction level. The hedonic and aspiration treadmills are likely to make inequalities in utility rather shallow and quite different from material inequalities. The most miserable are those who suffer from psychological problems, from great physical pain, or from unadjusted

aspiration levels, not necessarily the materially poor and the socially oppressed—even if a correlation between socio-economic status and SWB is clearly observed.⁹⁰

The liberal model would presumably advocate comparing people in some metric that would take account of their own judgments over the dimensions of life. For instance, Rawls suggests to rely on the preferences of a representative individual from the worst off group in order to weight the various primary goods. However, Rawls does affirm that society should focus on resources and opportunities, in the assessment of the distribution, rather than on whatever people care about in their lives. One can imagine an even more liberal approach that considers that, when individuals have been able to form consistent views of the good life, these views should provide the metric of evaluation. Such a super-liberal approach could entail quite different consequences depending on the values of the population. If the only dimension of life that people value were some kind of hedonic utility, then this approach would advocate comparing individuals in terms of hedonic utility. The super-liberal approach could then coincide with hedonic welfarism. If the only dimension of life valued by the population were wealth, then equalizing wealth would be the way to equalizing the value of life across individuals. If, as is more likely to be the case, people had different values and, therefore, different ordinal rankings of lives, one would have to find a way to reconcile their divergent evaluations.

⁸⁸ Reluctance to incorporate other-regarding preferences in welfare economics may have been exaggerated by the utilitarian perspective which, because of its additive structure, mechanically gives advantage to the selfish over the altruist. With inequality-averse criteria the impact of other-regarding preferences is much less strong. To illustrate this point, consider a two-agent model with a selfish agent whose total utility equals his self-centered utility, $U_1 = u_1$, while the other is altruist, $U_2 = \alpha u_2 + (1 - \alpha)u_1$, with $0 < \alpha < 1$. Equalizing their total utilities, $U_1 = U_2$, requires $u_1 = u_2$.

⁸⁹ It may seem that the average U -index proposed by Kahneman and Krueger (2006) naturally focuses on the

more miserable individuals because they contribute more to it and may also contribute more to its variations as their own index is more likely to vary. But, as almost all individuals have a positive U -index (in a survey, these authors found that people who are "very satisfied" with their life spend on average 11 percent of their time in an unpleasant state) no special weight is given to the plight of the more miserable if one simply computes the average U -index over the whole population. An inequality-averse social index is needed to give them more weight.

⁹⁰ Such a link is even observed for physical pain by Krueger and Arthur A. Stone (2008).

This is the indexing problem, which is examined in the next section.

In conclusion, the idea that GDP should be replaced by a single index of national utility is far from consensual, even among happiness specialists. This does not make happiness studies useless and a panel of subjective indicators would provide valuable information to complement more objective indicators of living conditions. A better connection between the recent happiness literature and earlier debates about liberalism and utilitarianism would be welcome because the happiness literature is replete with normative conclusions that do not seem sufficiently founded. Interestingly, the happiness studies can also prove useful for the liberal conception. Indeed, in spite of the aspiration treadmill, a clever use of satisfaction data, or the use of new questionnaires that would enable the respondents to express their ordinal preferences more directly than through the prism of a satisfaction level, may provide valuable information about people's preferences and values relative to the various dimensions of life (Clark and Oswald 2002; Schokkaert 2007a; Krueger 2008).

7. Capabilities

The capability approach has become an important reference in the field of alternative indicators. Developed by Sen (1985, 1992),⁹¹ it is often presented as intermediate between resourcist and welfarist approaches, but it is perhaps accurate to present it as more general. The two key terms in the approach are “functionings” and “capability sets” and they are defined as follows. “Functioning” is a catchword for any doing or being in the life of an individual, such as a consumption

bundle, a health condition, a level of education, and so on. At any moment in life, or over the whole life cycle, the actual situation of an individual can be described by a functioning vector. A “capability set” is the set of potential functioning vectors that an individual can obtain if he so chooses. While functionings describe achievements, capabilities capture opportunities. Sen proposed capabilities as the metric of advantage for the definition of a liberal egalitarian theory of justice. This proposal has attracted a lot of interest in particular because it makes it possible to take into account all the relevant dimensions of life, in contrast with the resourcist and hedonic approaches that can be criticized, in comparison, as relatively narrow.

7.1 Toward Applications

Being so general—what is not a “doing or being” in one's life?—the approach needs to be specified in order to inspire original applications. The body of empirical literature that takes inspiration from the capability approach is now numerically impressive. As noted in Robeyns (2006) and Schokkaert (2007b), in many cases the empirical studies are essentially similar, but for terminology, to the sociological studies of living conditions. But there are more original applications, e.g., when an evaluation of development programs that takes account of capabilities is contrasted with cost–benefit analysis (Sabina Alkire 2002), when a poverty rate is corrected in order to take account of the extra expenses borne by disabled persons (Kuklys 2005), or when a list of basic capabilities is enshrined in a theory of what a just society should provide to all citizens (Nussbaum 2000). More generally, all studies that seek to incorporate multiple dimensions of quality of life into the evaluation of individual and social situations can be considered, broadly speaking, as pertaining to this approach.

Two central questions pervade the empirical applications. The first concerns the

⁹¹ Recent surveys on this approach and its applications can be found in Kaushik Basu and Luis-Felipe Lopez-Calva (forthcoming), Wiebke Kuklys (2005), Ingrid Robeyns (2006), Robeyns and R. J. van der Veen (2007), Schokkaert (2007b).

distinction between capabilities and functionings. The latter are easier to observe because individual achievements are more accessible to the statistician than pure potentialities. There is also the normative issue of whether the evaluation of individual situations should be based on capabilities only, viewed as opportunity sets, or should take account of achieved functionings as well. The second central question is the index problem, which has also been raised about Rawls's theory of primary goods. There are many dimensions of functionings and capabilities and not all of them are equally valuable. The definition of a proper system of weights has appeared problematic in connection to the difficulties of social choice theory.

7.2 The Index Problem

The two questions are related. The indexing problem is often presented as a dilemma. If a single index is defined in order to weight the various dimensions of life for all individuals, uniformly, there is no chance that the weights will respect the individuals' own values about life and the procedure will appear paternalistic or perfectionist. If, on the contrary, one seeks to respect each and every individual's views about the relative importance of the various dimensions of life, one is taking the welfarist route, it is said, and ends up dealing with utility functions. In this light, turning attention to capability sets, as opposed to functionings, is often viewed as a solution. Even if the index of capabilities is the same objective index for all individuals, capabilities are just opportunity sets and individuals are free to choose from these sets, so that a society seeking to equalize capability indexes across individuals can still be considered a liberal society.⁹²

⁹² A slightly different index problem, namely, the case in which preferences are not consistent or stable, has also led Robert Sugden (2004, 2007) to propose opportunities as the relevant metric of well-being.

An additional possibility, also suggested by Sen, consists in abandoning the project of a precise numerical index and in devising a partial ordering of individual situations based on the intersection of individual orderings. Assuming that individuals have well-defined preference orderings over capability sets or functionings, one can consider that individual i is better off than individual j , in terms of capabilities or functionings, if this judgment is shared by all the preferences of the population (or all the preferences in a suitable domain). Rawls also invoked the possibility of this kind of unanimous judgment in order to identify the worst off group in terms of primary goods.

This "intersection" approach, however, has paradoxical consequences.⁹³ Suppose that individual situations are represented by vectors in some space of capabilities or functionings, and that all reasonable preferences are monotonic. In this context, the intersection approach can motivate a dominance principle stipulating that an individual having more in all dimensions is better off. All monotonic individual preferences do indeed agree with this principle. But another natural way to respect individual preferences, especially if one wants to avoid paternalism, is to say that two situations that are deemed equivalent by an individual should be considered equivalent when they are given to this individual. As it turns out, these two ideas clash, as we now explain. Formally, the index problem can be described as the search for a transitive but not necessarily complete ranking $\succeq$ of individual situations (x, i) , where x is some vector of individual functionings or capabilities and i is the name of the individual. The pair (x, i) says that i is in situation x . The expression $(x, i) \succeq (y, j)$ means that i in situation x is at least as well off as j in situation y . Individual preferences also need to be introduced and the expression

⁹³ See, e.g., Bernt Christian Brun and Bertil Tungodden (2004).

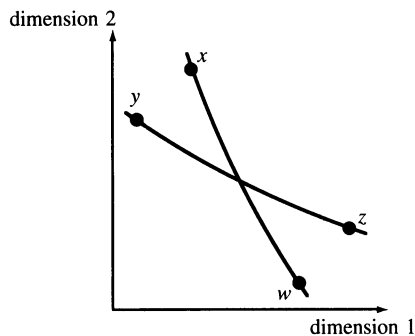


Figure 1. Dominance versus Nonpaternalism

$xR_i y$ means that x is at least as good as y for i 's preferences, while $xI_i y$ denotes indifference. Individual preferences are assumed to be monotonically increasing. The framework being defined, we can now formulate the ethical requirements introduced earlier in this paragraph. The *dominance principle* means that for all i, j , all x, y , if $x \gg y$ then $(x, i) \succ (y, j)$. The *nonpaternalism requirement* means that for all i , all x, y , if $xI_i y$ then $(x, i) \sim (y, i)$. Now, these two requirements are incompatible whenever there are individuals with different preferences. Because, under very mild assumptions which need not be detailed here, one can then find x, y, z, w and i, j such that $x \gg y, z \gg w, xI_i w$ and $yI_j z$ (see figure 1). By the dominance principle, $(x, i) \succ (y, j)$ and $(z, j) \succ (w, i)$, but by nonpaternalism, $(x, i) \sim (w, i)$ and $(y, j) \sim (z, j)$, which violates transitivity.

This fact implies that one must relax one of the two requirements. If one gives priority to nonpaternalism, one can nonetheless seek to preserve as much of the dominance principle as is compatible with nonpaternalism. A restricted dominance principle, for instance, stipulates that at least for some subset $A \subseteq X$, it holds that for all i, j , all $x, y \in A$, $x \gg y$ implies $(x, i) \succ (y, j)$. Under mild regularity conditions, the combination of this restricted dominance principle with nonpaternalism

implies that the ranking $\succeq$ must be of the egalitarian-equivalent sort.⁹⁴ In particular, the subset A over which dominance applies has to be a monotone path (an increasing curve) and interpersonal comparisons are performed as follows: $(x, i) \succeq (y, j)$ if and only if $a \geq b$, where $a, b \in A$ are defined by $xI_i a$ and $yI_j b$. The equivalent-income approach is an example of such a ranking, with a path defined by all levels of income associated with the reference values for the nonincome dimensions. This example shows that the indexing problem does not imply the alleged dilemma between paternalism and welfarism. It is possible to avoid welfarism while respecting preferences at the individual level.

If one instead gives priority to the dominance principle but seeks to avoid paternalism as much as possible, a natural solution is to take a reference preference ordering over X that is somehow a summary of the population preferences.⁹⁵

⁹⁴ See Fleurbaey (2007).

⁹⁵ Achin Chakraborty (1996) proposes to compute an "average" preference ordering for this purpose. Sprumont (2007) has also explored this kind of solution and given it an axiomatic foundation based on the dominance principle.

7.3 *Capabilities or Functionings?*⁹⁶

The other question that pervades the literature is whether one should focus on functionings or on capabilities. The empirical literature deals almost exclusively with functionings because they are observable. However, many functionings, such as income and health, also directly determine capabilities (to consume, to move around . . .). In addition, some recent work has explored the possibility to inquire directly by questionnaires whether individuals feel able to do certain things.⁹⁶

Independently of observability, there is the normative issue of whether the appropriate metric for interpersonal comparisons corresponds to achievements or to opportunities. Several justifications for an opportunity metric have been provided. One has already been mentioned, namely, the need to respect individual preferences without falling back into welfarism. But we have seen that the egalitarian-equivalent approach makes it possible, if one wishes, to focus on achievements while respecting individual preferences.

The main justification in Sen's work for a focus on capabilities is the importance of freedom. Classical welfare economics can indeed be criticized for being tied to the measurement of achievements (income, consumption, utility) and for being impervious to the importance of the freedom to choose. Several authors, however, have noted that the importance of freedom justifies taking account of capabilities, but not necessarily an exclusive focus on capabilities.⁹⁷ One can imagine a broader framework in which both capabilities and achieved functionings serve to describe individual situations and are weighted according to individuals' valuations of the relative importance of opportunity and achievement. This would come close

to a variant actually proposed by Sen (1992), which is formulated in terms of "refined functionings," i.e., the achieved functioning vectors combined with information about the capability sets from which they are obtained.

Another possible justification for an exclusive focus on opportunities comes from theories of justice based on personal responsibility. According to such theories, success or failure should be up to individuals provided they have been given suitable opportunities.⁹⁸ Several authors have criticized this approach for being potentially harsh with the losers and for embroiling distributive justice in the metaphysics of free will.⁹⁹ It has also been noted that opportunities are not always well defined in a social context because what is accessible to an individual may depend on what the others achieve. For instance, consumers are often momentarily rationed when others have bought all the local stock of a particular commodity. The possibility to have a successful career may be apparently open to two parents but, in fact, at least one of them must take care of the children.¹⁰⁰ A branch of the economic literature has explored how to measure opportunities and define reasonable criteria of equal opportunity as well as measures of opportunity inequalities.¹⁰¹

In conclusion, the capability approach has the merit of providing a lot of flexibility about the evaluation of individual situations, especially if one considers legitimate to take account of functionings as well. This flexibility comes at a cost, which is the difficulty for many authors to specify a precise measurement method. One can, however, consider that making progress on the indexing problem should help substantially in applications.

⁹⁸ See in particular Arneson (1989), G. A. Cohen (1989).

⁹⁹ See, e.g., Elizabeth S. Anderson (1999).

¹⁰⁰ See, e.g., Basu and Lopez-Calva (forthcoming).

¹⁰¹ An influential contribution is Roemer (1998). A recent discussion of this field can be found in Fleurbaey (2008).

⁹⁶ See, e.g., Paul Anand, Graham Hunter, and Ron Smith (2005).

⁹⁷ See, e.g., Richard J. Arneson (1998).

8. Concluding Remarks

Where do we stand in the quest for a better measure of social welfare? Let me briefly summarize my conclusions relative to the four alternatives to GDP introduced in the beginning of this paper. (1) The idea of computing a “corrected GDP” is definitely worth exploring but not by computing the total value of an extended vector of consumption with imputed prices for the nonmarket components because the link between such a magnitude and social welfare is problematic. In particular, green accounting, although based on an ingenious theory, does not seem practically superior to direct forecasts of the welfare of future generations. A more promising approach for the incorporation of non-monetary aspects of quality of life involves equivalent incomes in which income can be submitted to additions and subtractions reflecting people’s willingness-to-pay for a move from their current situation to a reference situation. One should, however, remain open to other possible indexing methods pertaining to the “equivalence” approach or even coming from different corners of the theory of fairness. (2) The idea of “*Gross National Happiness*” appears to be supplanted in the literature by the idea that separate data on the various components of SWB should be produced, with special care about the difficulties of comparing statements of satisfaction and happiness across populations. Special attention should also be devoted to the adaptation phenomenon, as one may be afraid that direct measures of SWB, reflecting how people feel rather than how well they live, do not reliably track material and social conditions. (3) The “*capability approach*” has inspired many interesting developments but consensus remains to be found on the index problem, a key issue in this respect being whether one should try to take account of preferences and values at the individual level. (4) “*Synthetic indicators*” appear inferior to

the other approaches because they do not seek to evaluate the social situation on the basis of the distribution of individual situations and also because they offer little clue for the choice of weights. However, the main advantage of such objective measures is that they are relatively simple to implement and are immune to the vagaries of people’s subjectivity. In spite of being inferior in theory to the individualistic approaches highlighted in this paper, social indicators of the HDI sort might remain valuable as long as the lack of data on individual situations in many countries hampers the application of more sophisticated approaches.

Let us now focus on the three more promising approaches, first examining further how they relate to each other, before examining prospects for future research and applications. What differentiates these approaches is not really the way in which they conceive of the aggregation of individual well-being into social welfare, because, although the literature on the equivalence and the capability approaches tends to advocate a strong inequality aversion while the literature on SWB often presents average numbers (such as the average satisfaction or the average *U*-index), this is a superficial difference that can be easily reduced by diversifying the range of aggregate indicators used in each approach. The key differences between the three approaches have to do with the evaluation of *individual* situations. In this respect, each approach can present itself as more comprehensive than the others. For instance, the capability approach can take subjective well-being as one functioning among the others and argue that it is open to all possible valuations of the dimensions of life. The equivalence approach can take a similar stance and, moreover, put forth its ability to be faithful to every individual’s evaluation of the relative importance of the various dimensions. The SWB approach can consider that it measures the ultimate outcome, while all the

objective aspects measured and weighted in the other two approaches are just inputs in the production of satisfaction or happiness.

A simple model may summarize these descriptions. Let the level of satisfaction with life be defined by a function $S(H, F)$, where H represents hedonic states and F the rest of the functionings. The SWB approach seeks to measure either S or H and related magnitudes (like the U -index, which is roughly¹⁰² the proportion of time with $H < 0$). The equivalence approach transforms the S function into a calibrated utility function $U(H, F)$ that is ordinally equivalent to S but is measured in the units of some functionings (such as income) and thereby eliminates the effects of the satisfaction treadmill. The capability approach emphasizes the opportunities for vectors (H, F) which are offered to the individuals, remaining rather shy about how to measure such magnitudes by an index.

Underlying the difference between the SWB approach and the other two is an old opposition between two important schools of thought. One is the *welfarist* approach, which is interested in promoting hedonic utility or satisfaction and has been a source of inspiration for many authors studying SWB. The other is the *liberal* approach, which seeks to give people what they value, reasoning in terms of resources or opportunities, and has motivated much of the theory of social choice and fair allocation as well as the capability approach. Elements of convergence can, however, be pointed out, which have to do with the prospects for future research and progress in the various approaches. Let us examine them in turn.

As far as SWB is concerned, progress will certainly be made in the future about issues that have already been of great concern, namely, the causal mechanisms relating

SWB to objective achievements and the strength of adaptation (the satisfaction and hedonic treadmills). An additional direction of research is suggested by the analysis proposed here. The distinction between affects and judgments being now firmly established in the happiness literature, it remains to clarify the interaction between the two with better data in which they are clearly distinguished. If one adopts the little model introduced two paragraphs above, it appears sensible to seek to estimate the relative importance that people assign to their hedonic states in the judgments they cast over their whole lives. Another important issue is individual heterogeneity. Most studies about the determinants of SWB are interested in some kind of average function of satisfaction or happiness, and it would be interesting to analyze individual diversity in more detail. These two questions are aimed at the same goal, namely, an estimation of the distribution of S functions—as opposed to satisfaction *levels*—in the population.

As far as the equivalence approach is concerned, progress on the side of applications would require similar data, namely, an estimation of the distribution of preferences (jointly with objective situations) in the population. A related issue is to assess how (im)precise and (in)consistent individual preferences are on the various aspects of personal situations. Note that imprecise individual preferences do not necessarily undermine the ethical appeal of the approach, as individuals would not object to evaluations of their own situations that are in the range of their possible preferences.¹⁰³ On the theory side, progress is needed on the elucidation of the ethical underpinnings of the various

¹⁰² Recall that it is the proportion of time in which the dominant affect is negative.

¹⁰³ One may also suspect that many apparent forms of irrationality are connected either to imprecise preferences or to the (rational) management of emotions—inconsistent rankings of F may be explained by consistent rankings of (H, F) .

possible choices of the reference situations that serve in the computation of "equivalent" sets or incomes. It is also possible that serious alternatives to the equivalence approach will emerge from the theory of fair allocation.

As far as the capability approach is concerned, it has already been explained in the previous section that the index problem is the key obstacle to more systematic applications. This is partly a theoretical and partly an empirical issue. Two schools of thought compete on the theory side. The liberal approach pushes in the direction of taking account of the concerned individuals' values and preferences, while a more perfectionist approach would be satisfied with weights derived from moral considerations.¹⁰⁴ One can suspect that the liberal approach has never seriously considered using indexes representing each individual's preferences for fear that this would amount to embracing welfarism. As we have seen in subsection 7.2, however, there is no such dilemma and the equivalence approach can offer itself as a possible index method.

The elements of convergence can then be described as follows. First, all approaches would be jointly interested in the empirical project of seeking a better understanding of the distribution of individual preferences and values. If the capability approach accepts to solve the index problem by seeking to respect each individual's views on his own situation,¹⁰⁵ the difference with the research line that inspires the equivalence approach more or less vanishes. If the SWB approach accepts that hedonic states are just an aspect

of life among others that may matter in individuals' satisfaction judgments and accepts that satisfaction judgments are ultimately more relevant for a complete evaluation of individual situations, the only difference that remains with the other two approaches is the treatment of the satisfaction treadmill. If the SWB approach proceeded to filter the effects of adaptation, the difference with the other two approaches might be substantially reduced.

A full convergence is not likely to occur soon. As the schools of thought just mentioned (welfarism, liberalism, perfectionism) are prominent in modern culture and public debates and will stay with us for a long time, it probably makes more sense to abandon the illusion that a unique consensual measure of well-being will emerge and to assign economists the task of thinking rigorously about the alternatives to GDP that each school of thought may inspire and of developing concrete proposals for each of them. As we have seen, such developments are under way. Serious alternatives to GDP are around the corner.

REFERENCES

- Alkire, Sabina. 2002. *Valuing Freedoms: Sen's Capability Approach and Poverty Reduction*. Oxford and New York: Oxford University Press.
- Anand, Paul, Graham Hunter, and Ron Smith. 2005. "Capabilities and Well-Being: Evidence Based on the Sen–Nussbaum Approach to Welfare." *Social Indicators Research*, 74(1): 9–55.
- Anderson, Elizabeth S. 1999. "What is the Point of Equality?" *Ethics*, 109(2): 287–337.
- Arneson, Richard J. 1989. "Equality and Equal Opportunity for Welfare." *Philosophical Studies*, 56(1): 77–93.
- Arneson, Richard J. 1998. "Real Freedom and Distributive Justice." In *Freedom in Economics: New Perspectives in Normative Analysis*, ed. Jean-François Laslier, Marc Fleurbaey, Nicolas Gravel, and Alain Trannoy, 165–96. London and New York: Routledge.
- Aronsson, Thomas, Per-Olov Johansson, and Karl-Gustaf Lofgren. 1997. *Welfare Measurement, Sustainability and Green National Accounting: A Growth Theoretical Approach*. Cheltenham, U.K. and Lyme, N.H.: Elgar.
- Arrow, Kenneth J. 1951. *Social Choice and Individual*

¹⁰⁴ A third view is that some dimensions of capabilities are incommensurable and cannot be put together in a synthetic index. It appears also somewhat illiberal if the concerned individuals are willing to make trade-offs between such dimensions in their own lives.

¹⁰⁵ This is not a minor step if one includes in it accepting that whether opportunities or achievements matter more should also be left to the concerned individuals' appreciation.

- Values. New York: Wiley.
- Arrow, Kenneth J., Partha S. Dasgupta, and Karl-Goran Mäler. 2003a. "Evaluating Projects and Assessing Sustainable Development in Imperfect Economies." *Environmental and Resource Economics*, 26(4): 647–85.
- Arrow, Kenneth J., Partha S. Dasgupta, and Karl-Goran Mäler. 2003b. "The Genuine Savings Criterion and the Value of Population." *Economic Theory*, 21(2–3): 217–25.
- Asheim, Geir B. 2000. "Green National Accounting: Why and How?" *Environment and Development Economics*, 5(1–2): 25–48.
- Asheim, Geir B. 2007a. "Can NNP Be Used for Welfare Comparisons?" *Environment and Development Economics*, 12(1): 11–31.
- Asheim, Geir B. 2007b. *Justifying, Characterizing and Indicating Sustainability*. Dordrecht: Springer.
- Asheim, Geir B., Wolfgang Buchholz, and Cees Withagen. 2003. "The Hartwick Rule: Myths and Facts." *Environmental and Resource Economics*, 25(2): 129–50.
- Asheim, Geir B., and Martin L. Weitzman. 2001. "Does NNP Growth Indicate Welfare Improvement?" *Economics Letters*, 73(2): 233–39.
- Atkinson, Anthony B. 1970. "On the Measurement of Inequality." *Journal of Economic Theory*, 2(3): 244–63.
- Atkinson, Anthony B., and Joseph E. Stiglitz. 1980. *Lectures on Public Economics*. London: McGraw-Hill.
- Barry, Brian. 2008. "Rationality and Want-Satisfaction." In *Justice, Political Liberalism, and Utilitarianism: Themes from Harsanyi and Rawls*, ed. Marc Fleurbaey, Maurice Salles, and John A. Weymark, 281–99. Cambridge and New York: Cambridge University Press.
- Basu, Kaushik, and Luis-Felipe Lopez-Calva. Forthcoming. "Functionings and Capabilities." In *Handbook of Social Choice and Welfare, Volume 2*, ed. Kenneth J. Arrow, Amartya K. Sen, and Kotaro Suzumura. Amsterdam and San Diego: Elsevier, North-Holland.
- Becker, Gary S., Tomas J. Philipson, and Rodrigo R. Soares. 2005. "The Quantity and Quality of Life and the Evolution of World Inequality." *American Economic Review*, 95(1): 277–91.
- Becker, Gary S., and Luis Rayo. 2008. "Economic Growth and Subjective Well-Being: Reassessing the Easterlin Paradox: Comment." *Brookings Papers on Economic Activity*, 1: 88–95.
- Bergson, Abram. 1938. "A Reformulation of Certain Aspects of Welfare Economics." *Quarterly Journal of Economics*, 52(2): 310–34.
- Bergson, Abram. 1966. *Essays in Normative Economics*. Cambridge and London: Harvard University Press.
- Bernheim, B. Douglas. 2008. "Behavioral Welfare Economics." National Bureau of Economic Research Working Paper 14622.
- Blackorby, Charles, Walter Bossert, and David Donaldson. 2005. *Population Issues in Social Choice Theory, Welfare Economics, and Ethics*. Cambridge and New York: Cambridge University Press.
- Blackorby, Charles, and David Donaldson. 1988. "Money Metric Utility: Harmless Normalization?" *Journal of Economic Theory*, 46(1): 120–29.
- Blackorby, Charles, and David Donaldson. 1990. "A Review Article: The Case against the Use of the Sum of Compensating Variations in Cost-Benefit Analysis." *Canadian Journal of Economics*, 23(3): 471–94.
- Boadway, Robin W., and Neil Bruce. 1984. *Welfare Economics*. Oxford: Basil Blackwell.
- Bommier, Antoine, and Stephane Zuber. 2008. "Can Preferences for Catastrophe Avoidance Reconcile Social Discounting with Intergenerational Equity?" *Social Choice and Welfare*, 31(3): 415–34.
- Brickman, Philip, and Donald T. Campbell. 1971. "Hedonic Relativism and Planning the Good Society." In *Adaptation-Level Theory: A Symposium*, ed. Mortimer H. Appley, 287–304. New York: Academic Press.
- Browning, Martin, Pierre-Andre Chiappori, and Arthur Lewbel. 2006. "Estimating Consumption Economies of Scale, Adult Equivalence Scales, and Household Bargaining Power." Boston College Department of Economics Working Paper 588.
- Brun, Bernt Christian, and Bertil Tungodden. 2004. "Non-welfaristic Theories of Justice: Is 'The Intersection Approach' a Solution to the Indexing Impasse?" *Social Choice and Welfare*, 22(1): 49–60.
- Burchardt, Tania. 2006. "Happiness and Social Policy: Barking Up the Right Tree in the Wrong Neck of the Woods." In *Social Policy Review: Analysis and Debate in Social Policy, 2006*, ed. Linda Bauld, Karen Clarke, and Tony Maltby, 145–64. Bristol: Policy Press.
- Chakraborty, Achin. 1996. "On the Possibility of a Weighting System for Functionings." *Indian Economic Review*, 31(2): 241–50.
- Clark, Andrew E., Paul Frijters, and Michael A. Shields. 2008. "Relative Income, Happiness, and Utility: An Explanation for the Easterlin Paradox and Other Puzzles." *Journal of Economic Literature*, 46(1): 95–144.
- Clark, Andrew E., and Andrew J. Oswald. 2002. "A Simple Statistical Method for Measuring How Life Events Affect Happiness." *International Journal of Epidemiology*, 31(6): 1139–44.
- Cohen, G. A. 1989. "On the Currency of Egalitarian Justice." *Ethics*, 99(4): 906–44.
- Dasgupta, Partha S., and Geoffrey M. Heal. 1979. *Economic Theory and Exhaustible Resources*. Cambridge and New York: Cambridge University Press.
- Dasgupta, Partha S., and Karl-Goran Mäler. 2000. "Net National Product, Wealth, and Social Well-Being." *Environment and Development Economics*, 5(1–2): 69–93.
- Deaton, Angus. 1979. "The Distance Function in Consumer Behaviour with Applications to Index Numbers and Optimal Taxation." *Review of Economic*

- Studies*, 46(3): 391–405.
- Deaton, Angus. 2008. "Income, Health, and Well-Being around the World: Evidence from the Gallup World Poll." *Journal of Economic Perspectives*, 22(2): 53–72.
- Deaton, Angus, and John Muellbauer. 1980. *Economics and Consumer Behavior*. Cambridge and New York: Cambridge University Press.
- Debreu, Gerard. 1951. "The Coefficient of Resource Utilization." *Econometrica*, 19(3): 273–92.
- Diamond, Peter A., and Jerry A. Hausman. 1994. "Contingent Valuation: Is Some Number Better than No Number?" *Journal of Economic Perspectives*, 8(4): 45–64.
- Diener, Ed. 1994. "Assessing Subjective Well-Being: Progress and Opportunities." *Social Indicators Research*, 31(2): 103–57.
- Diener, Ed. 2000. "Subjective Well-Being: The Science of Happiness and a Proposal for a National Index." *American Psychologist*, 55(1): 34–43.
- Diener, Ed, Eunkook M. Suh, Richard E. Lucas, and Heidi L. Smith. 1999. "Subjective Well-Being: Three Decades of Progress." *Psychological Bulletin*, 125(2): 276–302.
- Diewert, W. E. 1976. "Exact and Superlative Index Numbers." *Journal of Econometrics*, 4(2): 115–45.
- Diewert, W. E. 1992a. "Exact and Superlative Welfare Change Indicators." *Economic Inquiry*, 30(4): 562–82.
- Diewert, W. E. 1992b. "Fisher Ideal Output, Input, and Productivity Indexes Revisited." *Journal of Productivity Analysis*, 3(3): 211–48.
- Dowrick, Steve, Yvonne Dunlop, and John Quiggin. 2003. "Social Indicators and Comparisons of Living Standards." *Journal of Development Economics*, 70(2): 501–29.
- Drèze, Jean, and Nicholas Stern. 1987. "The Theory of Cost-Benefit Analysis." In *Handbook of Public Economics, Volume 2*, ed. Alan J. Auerbach and Martin Feldstein, 909–90. Amsterdam: Elsevier, North-Holland.
- Dutta, Indranil, Prasanta K. Pattanaik, and Yongsheng Xu. 2003. "On Measuring Deprivation and the Standard of Living in a Multidimensional Framework on the Basis of Aggregate Data." *Economica*, 70(278): 197–221.
- Dworkin, Ronald. 2000. *Sovereign Virtue: The Theory and Practice of Equality*. Cambridge and London: Harvard University Press.
- Easterlin, Richard A. 1974. "Does Economic Growth Improve the Human Lot? Some Empirical Evidence." In *Nations and Households in Economic Growth: Essays in Honor of Moses Abramovitz*, ed. Paul A. David and Melvin W. Reder, 89–125. New York: Academic Press.
- Easterlin, Richard A. 1995. "Will Raising the Incomes of All Increase the Happiness of All?" *Journal of Economic Behavior and Organization*, 27(1): 35–47.
- Ekman, Paul. 1992. "An Argument for Basic Emotions." *Cognition and Emotion*, 6(3–4): 169–200.
- Fehr, Ernst, and Klaus M. Schmidt. 2003. "Theories of Fairness and Reciprocity: Evidence and Economic Applications." In *Advances in Economics and Econometrics: Theory and Applications, Eighth World Congress, Volume 1*, ed. Mathias Dewatripont, Lars Peter Hansen, and Stephen J. Turnovsky, 208–57. Cambridge and New York: Cambridge University Press.
- Feldstein, Martin S. 1972a. "Distributional Equity and the Optimal Structure of Public Prices." *American Economic Review*, 62(1): 32–36.
- Feldstein, Martin S. 1972b. "Equity and Efficiency in Public Pricing." *Quarterly Journal of Economics*, 86(2): 175–87.
- Ferrer-i-Carbonell, Ada, and Paul Frijters. 2004. "How Important Is Methodology for the Estimates of the Determinants of Happiness?" *Economic Journal*, 114(497): 641–59.
- Fisher, Irving. 1922. *The Making of Index Numbers: A Study of Their Varieties, Tests, and Reliability*. Boston: Houghton Mifflin.
- Fleurbaey, Marc. 2005. "Health, Wealth, and Fairness." *Journal of Public Economic Theory*, 7(2): 253–84.
- Fleurbaey, Marc. 2007. "Social Choice and the Indexing Dilemma." *Social Choice and Welfare*, 29(4): 633–48.
- Fleurbaey, Marc. 2008. *Fairness, Responsibility, and Welfare*. Oxford and New York: Oxford University Press.
- Fleurbaey, Marc, and Guillaume Gaulier. Forthcoming. "International Comparisons of Living Standards By Equivalent Incomes." *Scandinavian Journal of Economics*.
- Fleurbaey, Marc, and Francois Maniquet. Forthcoming. *A Theory of Fairness and Social Welfare*. Cambridge and New York: Cambridge University Press.
- Fleurbaey, Marc, and Philippe Mongin. 2005. "The News of the Death of Welfare Economics Is Greatly Exaggerated." *Social Choice and Welfare*, 25(2–3): 381–418.
- Frederick, Shane, and George Loewenstein. 1999. "Hedonic Adaptation." In *Well-Being: The Foundations of Hedonic Psychology*, ed. Daniel Kahneman, Ed Diener, and Norbert Schwarz, 302–29. New York: Russell Sage Foundation.
- Freeman, A. Myrick, III. 1993. *The Measurement of Environmental and Resource Values: Theory and Methods*. Washington, D.C.: Resources for the Future.
- Frey, Bruno S., and Alois Stutzer. 2002. *Happiness and Economics: How the Economy and Institutions Affect Well-Being*. Princeton and Oxford: Princeton University Press.
- Frey, Bruno S., and Alois Stutzer. 2007. "Should National Happiness Be Maximized?" University of Zurich Institute for Empirical Research in Economics Working Paper 306.
- Frijda, Nico H. 1999. "Emotions and Hedonic Experience." In *Well-Being: The Foundations of Hedonic Psychology*, ed. Daniel Kahneman, Ed Diener, and Norbert Schwarz, 190–212. New York: Russell Sage Foundation.

- Gadrey, Jean, and Florence Jany-Catrice. 2006. *The New Indicators of Well-Being and Development*. Houndmills, U.K. and New York: Palgrave Macmillan.
- Graaff, J. de V. 1977. "Equity and Efficiency as Components of the General Welfare." *South African Journal of Economics*, 45(4): 362–75.
- Grewal, Ini, Jane Lewis, Terry Flynn, Jackie Brown, John Bond, and Joanna Coast. 2006. "Developing Attributes for a Generic Quality of Life Measure for Older People: Preferences or Capabilities?" *Social Science and Medicine*, 62(8): 1891–1901.
- Hamilton, Kirk, and Michael Clemens. 1999. "Genuine Savings Rates in Developing Countries." *World Bank Economic Review*, 13(2): 333–56.
- Hammond, Peter J. 1978. "Economic Welfare with Rank Order Price Weighting." *Review of Economic Studies*, 45(2): 381–84.
- Hammond, Peter J. 1994. "Money Metric Measures of Individual and Social Welfare Allowing for Environmental Externalities." In *Models and Measurement of Welfare and Inequality*, ed. Wolfgang Eichhorn, 694–724. Berlin and Heidelberg: Springer.
- Hartwick, John M. 1977. "Intergenerational Equity and the Investing of Rents from Exhaustible Resources." *American Economic Review*, 67(5): 972–74.
- Hausman, Daniel M. 2007. "Hedonist Welfare Economics and the Value of Health." Unpublished.
- Heal, Geoffrey M., and Bengt Kriström. 2005. "National Income and the Environment." In *Handbook of Environmental Economics, Volume 3, Economywide and International Environmental Issues*, ed. Karl-Göran Mäler and Jeffrey R. Vincent, 1147–1217. Amsterdam and San Diego: Elsevier, North-Holland.
- Heal, Geoffrey M., and Bengt Kriström. 2008. "A Note on National Income in a Dynamic Economy." *Economics Letters*, 98(1): 2–8.
- Jorgenson, Dale W. 1990. "Aggregate Consumer Behavior and the Measurement of Social Welfare." *Econometrica*, 58(5): 1007–40.
- Jorgenson, Dale W. 1997. *Welfare, Volume 2, Measuring Social Welfare*. Cambridge and London: MIT Press.
- Kahneman, Daniel. 1999. "Objective Happiness." In *Well-Being: The Foundations of Hedonic Psychology*, ed. Daniel Kahneman, Ed Diener, and Norbert Schwarz, 3–25. New York: Russell Sage Foundation.
- Kahneman, Daniel, and Alan B. Krueger. 2006. "Developments in the Measurement of Subjective Well-Being." *Journal of Economic Perspectives*, 20(1): 3–24.
- Kahneman, Daniel, Alan B. Krueger, David A. Schkade, Norbert Schwarz, and Arthur A. Stone. 2004. "Toward National Well-Being Accounts." *American Economic Review*, 94(2): 429–34.
- Kahneman, Daniel, Ilana Ritov, and David A. Schkade. 1999. "Economic Preferences or Attitude Expressions?: An Analysis of Dollar Responses to Public Issues." *Journal of Risk and Uncertainty*, 19(1–3): 203–35.
- Kahneman, Daniel, Peter P. Wakker, and Rakesh Sarin. 1997. "Back to Bentham? Explorations of Experienced Utility." *Quarterly Journal of Economics*, 112(2): 375–405.
- King, Mervyn A. 1983. "Welfare Analysis of Tax Reforms Using Household Data." *Journal of Public Economics*, 21(2): 183–214.
- Kirman, Alan P. 1992. "Whom or What Does the Representative Individual Represent?" *Journal of Economic Perspectives*, 6(2): 117–36.
- Kolm, Serge-Christophe. 1968. "The Optimal Production of Social Justice." In *Economie Publique*, ed. H. Guitton and J. Margolis, 145–200. Paris: CNRS.
- Krueger, Alan B. 2008. "Economic Growth and Subjective Well-Being: Reassessing the Easterlin Paradox: Comment." *Brookings Papers on Economic Activity*, 1: 95–100.
- Krueger, Alan B., Daniel Kahneman, David A. Schkade, Norbert Schwarz, and Arthur A. Stone. 2008. "National Time Accounting: The Currency of Life." Princeton University Industrial Relations Section Working Paper 523.
- Krueger, Alan B., and David A. Schkade. 2008. "The Reliability of Subjective Well-Being Measures." *Journal of Public Economics*, 92(8–9): 1833–45.
- Krueger, Alan B., and Arthur A. Stone. 2008. "Assessment of pain: A Community-Based Diary Survey in the USA." *Lancet*, 371(9623): 1519–25.
- Kuklys, Wiebke. 2005. *Amartya Sen's Capability Approach: Theoretical Insights and Empirical Applications*. Berlin and New York: Springer.
- Lauwers, L. 2006. "Ordering Infinite Utility Streams: Completeness at the Cost of a Non-Ramsey Set." Unpublished.
- Layard, Richard. 2005. *Happiness: Lessons from a New Science*. London: Allen Lane.
- Layard, Richard, and Stephen Glaister, eds. 1994. *Cost-Benefit Analysis*, Second edition. Cambridge and New York: Cambridge University Press.
- Lewbel, Arthur. 1997. "Consumer Demand Systems and Household Equivalence Scales." In *Handbook of Applied Econometrics, Volume 2: Microeconomics*, ed. M. Hashem Pesaran and Peter Schmidt, 167–201. Malden, Mass. and Oxford: Blackwell.
- Loewenstein, George, and Peter A. Ubel. 2008. "Hedonic Adaptation and the Role of Decision and Experience Utility in Public Policy." *Journal of Public Economics*, 92(8–9): 1795–1810.
- Malmquist, Sten. 1953. "Index Numbers and Indifference Surfaces." *Trabajos de Estadística y de Investigación Operativa*, 4(2): 209–42.
- Maniquet, Francois. 2007. "Social Orderings and the Evaluation of Public Policy." *Revue d'Economie Politique*, 117(1): 37–60.
- Maniquet, Francois, and Yves Sprumont. 2004. "Fair Production and Allocation of an Excludable Nonrival Good." *Econometrica*, 72(2): 627–40.
- Miringoff, Marc, and Marque-Luisa Miringoff. 1999. *The Social Health of the Nation: How America Is Really Doing*. Oxford and New York: Oxford University Press.

- Moulin, Hervé, and William Thomson. 1997. "Axiomatic Analysis of Resource Allocation Problems." In *Social Choice Re-examined, Volume 1, Proceedings of the IEA Conference Held at Schloss Hernstein, Berndorf, near Vienna, Austria*, ed. Kenneth J. Arrow, Amartya Sen and Kotaro Suzumura, 101–20. New York: St. Martin's Press; London: Macmillan Press.
- Neumayer, Eric. 2004. "Sustainability and Well-Being Indicators." United Nations University World Institute for Development Economics Research Paper 2004/23.
- Ng, Yew-Kwang. 1986. "Social Criteria for Evaluating Population Change: An Alternative to the Blackorby–Donaldson Criterion." *Journal of Public Economics*, 29(3): 375–81.
- Nordhaus, William D., and James Tobin. 1973. "Is Growth Obsolete?" In *The Measurement of Economic and Social Performance: Studies in Income and Wealth, Volume 38*, ed. Milton Moss, 509–31. New York: National Bureau of Economic Research.
- Nussbaum, Martha C. 2000. *Women and Human Development: The Capabilities Approach*. Cambridge; New York and Melbourne: Cambridge University Press.
- Nussbaum, Martha C. 2008. "Who Is the Happy Warrior? Philosophy Poses Questions to Psychology." *Journal of Legal Studies*, 37(S2): S81–114.
- Osberg, Lars, and Andrew Sharpe. 2002. "An Index of Economic Well-Being for Selected OECD Countries." *Review of Income and Wealth*, 48(3): 291–316.
- Oswald, Andrew J. 1997. "Happiness and Economic Performance." *Economic Journal*, 107(445): 1815–31.
- Pazner, Elisha. 1979. "Equity, Nonfeasible Alternatives and Social Choice: A Reconsideration of the Concept of Social Welfare." In *Aggregation and Revelation of Preferences*, ed. Jean-Jacques Laffont, 161–73. Amsterdam and New York: Elsevier, North-Holland.
- Pollak, Robert A. 1981. "The Social Cost of Living Index." *Journal of Public Economics*, 15(3): 311–36.
- Ramsey, F. P. 1928. "A Mathematical Theory of Saving." *Economic Journal*, 38(152): 543–59.
- Ravallion, Martin. 1997. "Good and Bad Growth: The Human Development Reports." *World Development*, 25(5): 631–38.
- Rawls, John. 1971. *A Theory of Justice*. Cambridge and London: Harvard University Press.
- Rawls, John. 1982. "Social Unity and Primary Goods." In *Utilitarianism and Beyond*, ed. Amartya Sen and Bernard Williams, 159–86. Cambridge and New York: Cambridge University Press.
- Roberts, Kevin. 1980. "Price-Independent Welfare Prescriptions." *Empirical Economics*, 13(3): 277–97.
- Robeyns, Ingrid. 2006. "The Capability Approach in Practice." *Journal of Political Philosophy*, 14(3): 351–76.
- Robeyns, Ingrid, and R. J. van der Veen. 2007. "Sustainable Quality of Life: Conceptual Analysis for a Policy-Relevant Empirical Specification." Netherlands Environmental Assessment Agency Report 550031006/2007.
- Roemer, John E. 1998. *Equality of Opportunity*. Cambridge and London: Harvard University Press.
- Roemer, John E., and Kotaro Suzumura, eds. 2007. *Intergenerational Equity and Sustainability*. Houndmills, U.K. and New York: Palgrave Macmillan.
- Russell, James A. 1980. "A Circumplex Model of Affect." *Journal of Personality and Social Psychology*, 39(6): 1161–78.
- Samuelson, Paul A. 1947. *Foundations of Economic Analysis*. Cambridge and London: Harvard University Press.
- Samuelson, Paul A. 1956. "Social Indifference Curves." *Quarterly Journal of Economics*, 70(1): 1–22.
- Samuelson, Paul A. 1961. "The Evaluation of 'Social Income': Capital Formation and Wealth." In *The Theory of Capital*, ed. Friedrich A. Lutz and Douglas C. Hague, 32–57. London: Macmillan.
- Samuelson, Paul A. 1977. "Reaffirming the Existence of 'Reasonable' Bergson–Samuelson Social Welfare Functions." *Economica*, 44(173): 81–88.
- Samuelson, Paul A., and S. Swamy. 1974. "Invariant Economic Index Numbers and Canonical Duality: Survey and Synthesis." *American Economic Review*, 64(4): 566–93.
- Schokkaert, Erik. 2007a. "Capabilities and Satisfaction with Life." *Journal of Human Development*, 8(3): 415–30.
- Schokkaert, Erik. 2007b. "The Capabilities Approach." Catholic University Leuven Center for Economic Studies Discussion Paper 07.34.
- Sen, Amartya. 1970. *Collective Choice and Social Welfare*. San Francisco: Holden-Day.
- Sen, Amartya. 1976. "Real National Income." *Review of Economic Studies*, 43(1): 19–39.
- Sen, Amartya. 1979. "The Welfare Basis of Real Income Comparisons: A Survey." *Journal of Economic Literature*, 17(1): 1–45.
- Sen, Amartya. 1985. *Commodities and Capabilities*. Amsterdam: Elsevier, North-Holland.
- Sen, Amartya. 1992. *Inequality Reexamined*. New York: Russell Sage Foundation; Cambridge: Harvard University Press.
- Sen, Amartya. 1999. "The Possibility of Social Choice." *American Economic Review*, 89(3): 349–78.
- Sidgwick, Henry. 1907. *The Methods of Ethics*, Seventh edition. London: Macmillan.
- Slesnick, Daniel T. 1991. "Aggregate Deadweight Loss and Money Metric Social Welfare." *International Economic Review*, 32(1): 123–46.
- Slesnick, Daniel T. 1998. "Empirical Approaches to the Measurement of Welfare." *Journal of Economic Literature*, 36(4): 2108–65.
- Slesnick, Daniel T. 2001. *Consumption and Social Welfare: Living Standards and Their Distribution in the United States*. Cambridge; New York and Melbourne: Cambridge University Press.
- Sobel, Joel. 2005. "Interdependent Preferences and Reciprocity." *Journal of Economic Literature*, 43(2): 392–436.
- Sprumont, Yves. 2007. "Resource Egalitarianism with a Dash of Efficiency." Université de Montreal

- Département de sciences économique 2007-03.
- Stevenson, Betsey, and Justin Wolfers. 2008. "Economic Growth and Subjective Well-Being: Reassessing the Easterlin Paradox." *Brookings Papers on Economic Activity*, 1: 1–87.
- Sugden, Robert. 2004. "The Opportunity Criterion: Consumer Sovereignty without the Assumption of Coherent Preferences." *American Economic Review*, 94(4): 1014–33.
- Sugden, Robert. 2007. "The Value of Opportunities over Time When Preferences Are Unstable." *Social Choice and Welfare*, 29(4): 665–82.
- Van Praag, Bernard, and Ada Ferrer-i-Carbonell. 2007. *Happiness Quantified: A Satisfaction Calculus Approach*. Oxford and New York: Oxford University Press.
- Weitzman, Martin L. 1976. "On the Welfare Significance of National Product in a Dynamic Economy." *Quarterly Journal of Economics*, 90(1): 156–62.
- Weitzman, Martin L. 1988. "Consumer's Surplus as an Exact Approximation When Prices Are Appropriately Deflated." *Quarterly Journal of Economics*, 103(3): 543–53.
- Weitzman, Martin L. 1998. "On the Welfare Significance of National Product under Interest-Rate Uncertainty." *European Economic Review*, 42(8): 1581–94.
- Weitzman, Martin L. 2001. "A Contribution to the Theory of Welfare Accounting." *Scandinavian Journal of Economics*, 103(1): 1–23.
- Zame, William R. 2007. "Can Intergenerational Equity Be Operationalized?" *Theoretical Economics*, 2(2): 187–202.